

Luminite®

GRAVITON

—— midi controller ——

User Manual

For use with Graviton M1/M2 Software 5.3.x

www.luminitefx.com

Contents

Chapter A: Introduction

- Graviton M1
- Graviton M2

Chapter B: Basic Workflow

- Step 1: Connect & Pair
- Step 2: Programming Buttons
- Step 3: Programming Presets
- Step 4: Programming Expression / EC Controllers

Chapter C: Interface Walkthrough

- Touch User Interface
- Bank Mode / Song Mode
- Home Views
 - Preset Editor
 - Preset Library
 - Tracking
 - Live
- Songs Views
 - Playing Song
 - Playing Setlist
 - Song Library
 - Setlist Library
- Control Views
 - Button Settings
 - XY Controller
 - Footswitch Pedal
 - MIDI In
 - EC Controller
 - Expression Pedal
- Settings
 - General
 - Song Mode
 - MIDI Routing
 - Channel Filter
 - MIDI Monitor
- Safe Mode / Boot Issue

Chapter D: Glossary

Chapter E: Features & Specifications

Chapter A: Introduction

Graviton M1/M2 are next-generation MIDI controller pedals designed for guitarists and bassists who want to bring the full power of MIDI into their live or studio rigs — with total wireless freedom, effortless usability, compatibility, and expandability in an ultra-compact package.

Total Wireless Freedom

The **Graviton M1/M2** can be fully remote-controlled using **guitar-mounted XY/EC Controllers**, each programmable for sending MIDI commands, switching banks, and more. Their intuitive design allows **eyes-free operation**, much like using a pickup selector switch—eliminating the need for “tap dancing” and letting you move freely on stage.

F6W or **F10W wireless footswitches** can be placed anywhere—on stage or in the studio—for maximum flexibility. Built-in **Bluetooth MIDI** adds wireless connectivity to a wide range of MIDI devices.

Effortless Usability

The Graviton M1/M2 have a large **3.5-inch touchscreen LCD**, making MIDI programming fast and intuitive—**no PC editor required**. The user-friendly interface supports quick adjustments both on and off stage, and **automatically saves** every on-screen change. A wide range of MIDI commands and a suite of practical software features streamline your workflow.

Compatibility & Expandability

Control all types of gear—simultaneously—using **5-pin MIDI**, **TRS MIDI**, **USB MIDI**, and **Bluetooth MIDI**. Control amps and effects with **Relay switching**, send **Control Voltage** to synths, and sync everything to tempo with **MIDI Clock** or **Analog Clock**. Pair up to **10 XY/EC Controllers** (the 10 slots are shared between XY and EC) plus **10 wireless footswitches** (F6W/F10W), and connect virtually **any third-party footswitch or expression pedal** via the TRS jacks—**all working together seamlessly**. The **M1/M2** can also be **remotely controlled by incoming MIDI messages**, giving you **maximum flexibility** for any setup.

Ultra-Compact Footprint

The **Graviton M1** and **M2** are very compact, saving pedalboard space and weight. Use the M1/M2 to turn compact multi-effects pedals, amp modelers, computers, or iOS devices into **portable and powerful gigging machines**.

For the full feature list and hardware specs, see [Chapter E: Features & Specifications](#).

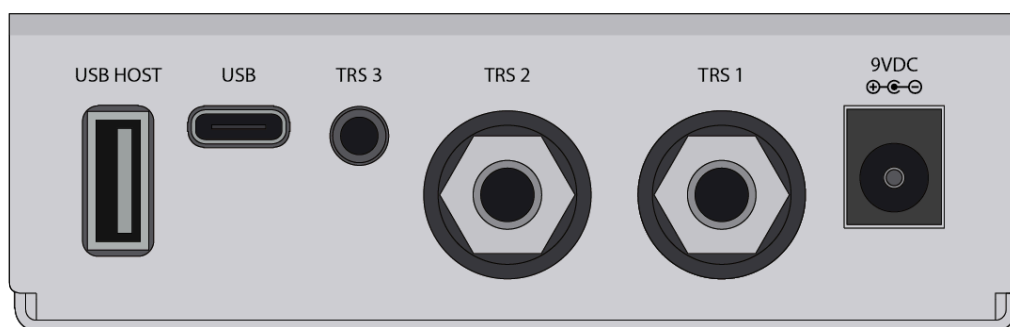
Graviton M1

Front



- **Display:** 3.5-inch touchscreen LCD.

Back

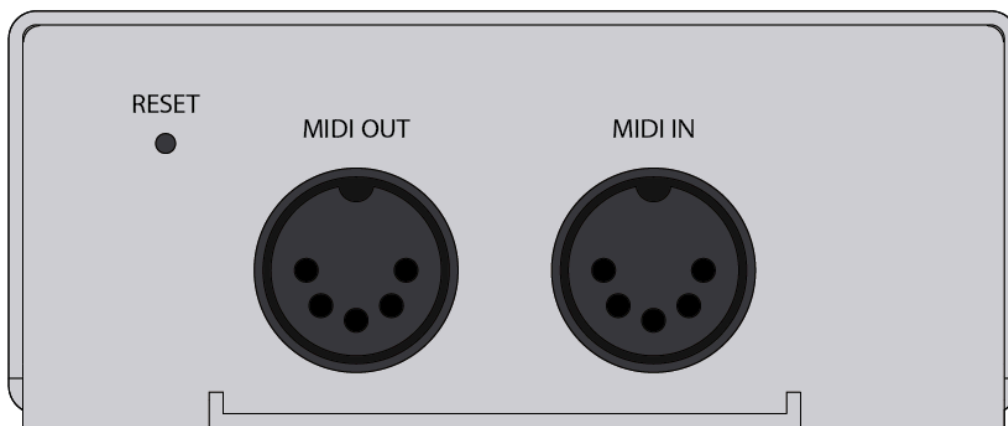


- **9VDC:** Power adapter input (9V center-negative, 500mA minimum).

- **TRS 1 and TRS 2:** MIDI Out (4 modes), or footswitch / expression pedal input, with 1/4-inch (6.35mm) stereo cables.
- **TRS 3 (1/8 Out): MIDI Out Only** — 1/8-inch (3.5mm) stereo MIDI out (Type A / Type B / Tip Active / Ring Active). No footswitch or expression pedal on this jack.
- **USB:** MIDI to and from a computer over a Type-C cable (the M1 can be USB bus powered).
- **USB Host:** MIDI to and from other USB MIDI devices. It can also power other devices, or charge **XY/EC Controllers** and **F6W/F10W** footswitches.

⚠ **Note:*** Use either **USB MIDI** or* *USB Host MIDI* — not both at the same time.

Side



- **MIDI In:** Standard 5-pin MIDI cable.
- **MIDI Out:** Standard 5-pin MIDI cable.
- **Factory Reset:** Insert a pin into the hole, then press and hold the button while powering on until the logo appears, then release. The device returns to its factory settings.

⚠ **Note:*** **A normal reset** (Settings → Data → Reset Data → All*) *is enough in most cases, and keeps your Backups. A factory reset also erases Backups — use it only if a normal reset doesn't fix the issue.*

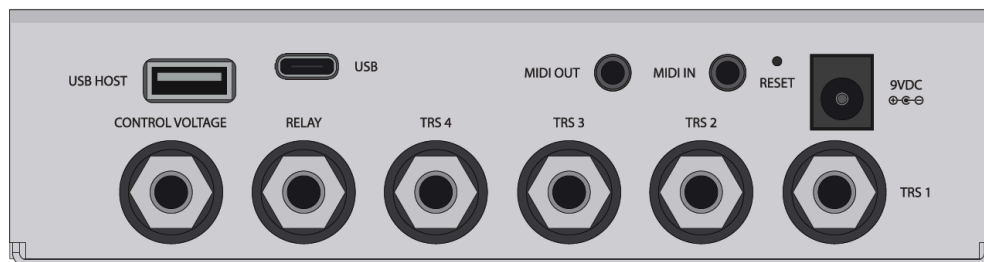
Graviton M2

Front



- **Display:** 3.5-inch touchscreen LCD.
- **6 × soft-touch footswitches:** Smooth and silent operation.
- **Staggered footswitch height:** Taller at the back and lower at the front for easy access.

Back

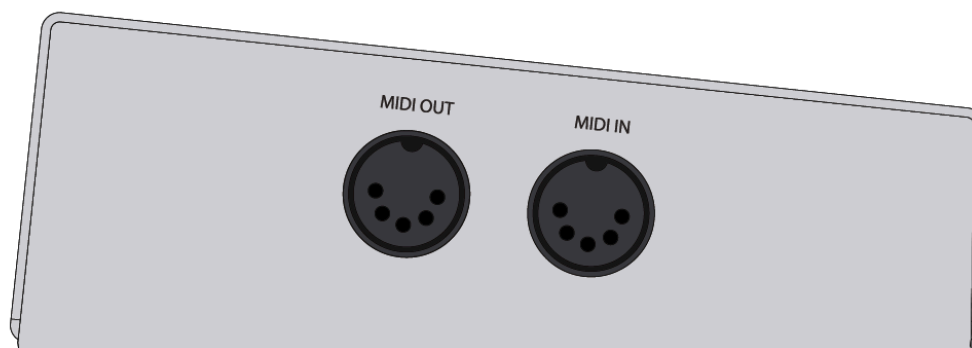


- **9VDC:** Power adapter input (9V center-negative, 500mA minimum).
- **TRS 1 to TRS 4:** MIDI Out (4 modes), or footswitch / expression pedal input, with 1/4-inch (6.35mm) stereo cables.
- **TRS 5 (1/8 Out): MIDI Out Only** — 1/8-inch (3.5mm) stereo MIDI out (Type A / Type B / Tip Active / Ring Active). No footswitch or expression pedal on this jack.

- **MIDI In (1/8):** 1/8-inch (3.5mm) stereo MIDI in, Type A only.
- **Relay:** Switch amp channels or pedal footswitch inputs remotely, with a 1/4-inch (6.35mm) stereo cable.
- **Control Voltage:** Send control voltage to CV inputs, or drive the expression inputs of synths and pedals, with a 1/4-inch (6.35mm) stereo cable.
- **USB:** MIDI to and from a computer over a Type-C cable (the M2 can be USB bus powered).
- **USB Host:** MIDI to and from other USB MIDI devices. It can also power other devices, or charge **XY/EC Controllers** and **F6W/F10W** footswitches.

⚠ **Note:*** Use either **USB MIDI** or* *USB Host MIDI* — not both at the same time.

Side



- **MIDI In:** Standard 5-pin MIDI cable.
- **MIDI Out:** Standard 5-pin MIDI cable.
- **Factory Reset:** Insert a pin into the hole, then press and hold the button while powering on until the logo appears, then release. The device returns to its factory settings.

⚠ **Note:*** Use either **the 1/8-inch MIDI In** or* *the 5-pin MIDI In* — not both at the same time.

⚠ **Note:*** A normal reset (Settings → Data → Reset Data → All*) is enough in most cases, and keeps your Backups. A factory reset also erases Backups — use it only if a normal reset doesn't fix the issue.

Chapter B: Basic Workflow

The workflow can be summarized in 4 steps:

Step 1. Connect & Pair: Pair your F10W, XY, or EC, plug in an Expression Pedal, and connect cables to your other MIDI devices (amp, modeler, etc.).

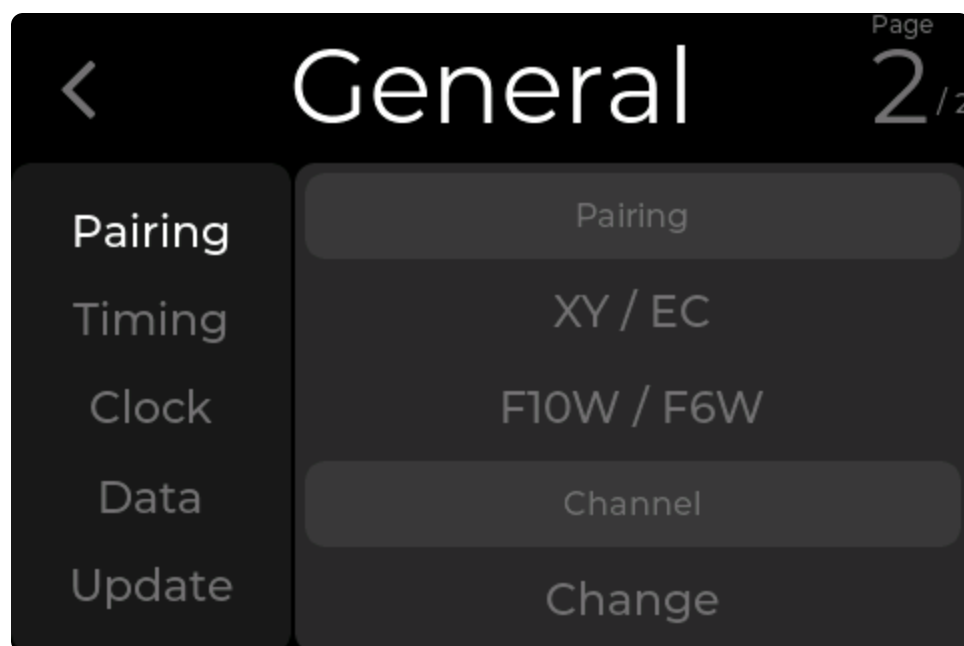
Step 2. Programming Buttons: What does this button do?

Step 3. Programming Presets: What MIDI messages are sent?

Step 4. Programming Expression / EC Controllers: What happens when you sweep the pedal or rotate this knob?

Step 1: Connect & Pair

Pairing a wireless controller, and plugging in TRS footswitches or an Expression Pedal, is all done under **Settings** → **General**. See the **Settings** chapter for details.



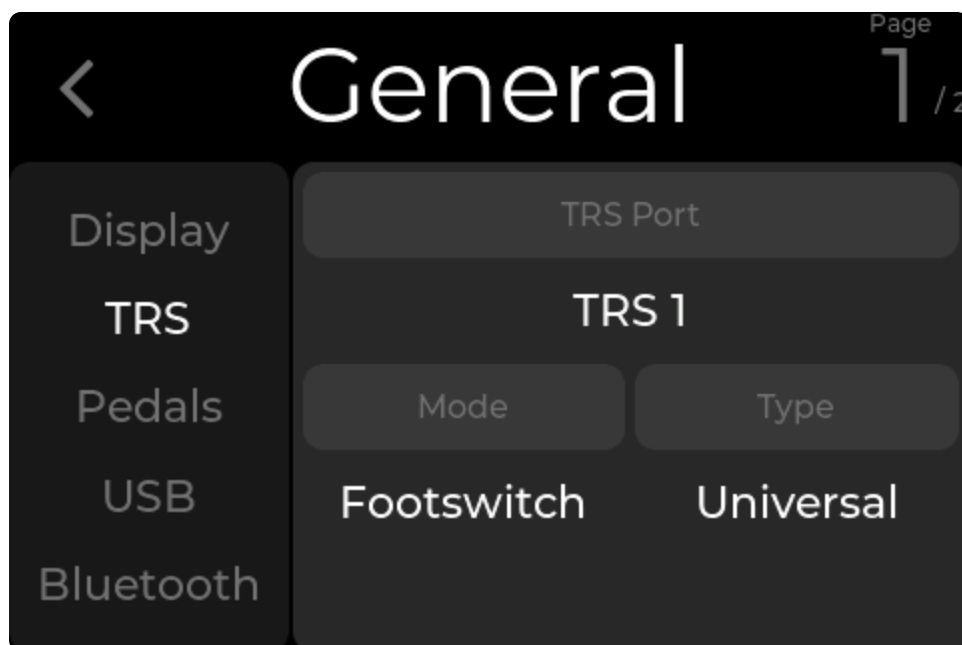
XY and EC Controllers

To pair: **General** > **Pairing** → **XY / EC** → tap an empty slot and follow the on-screen instructions.

Wireless Footswitch Pedals

To pair: **General** > **Pairing** → **F10W / F6W** → tap an empty slot and follow the on-screen instructions.

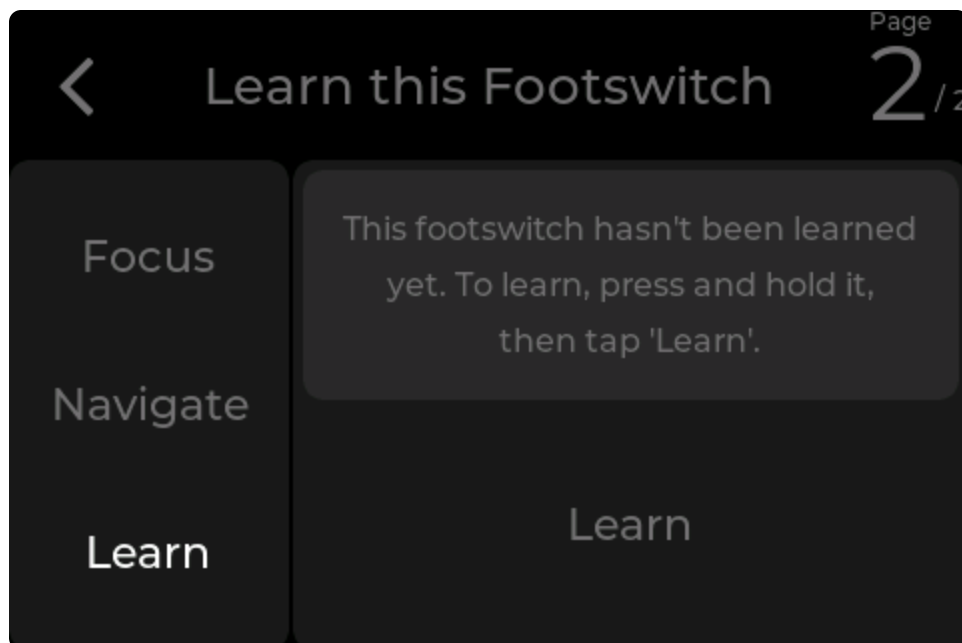
Wired Footswitch Pedals



1. Plug into a **TRS** jack with a 1/4" cable.
2. **General** > **TRS** → that jack → **Mode** → **Footswitch**. - **Type**: → **1/2/3-button, F10/W / F6/W** (Wired Mode) - Or **Universal** (custom / odd ladders — use **Learn**, below).

Debounce and calibration live under **General** > **Pedals**

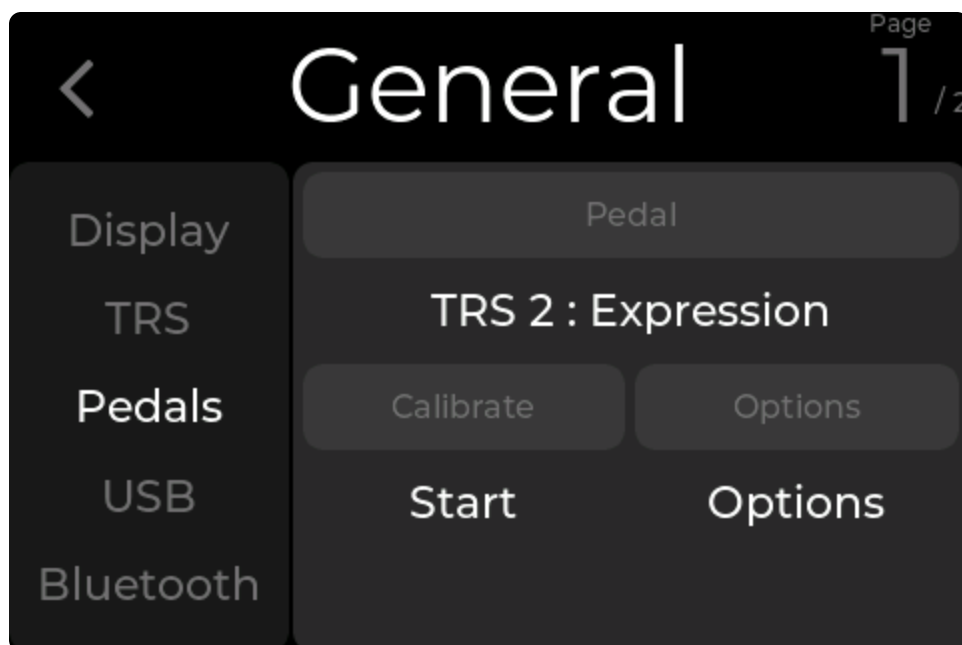
Universal Footswitch Learn



For **Universal** (UFS) only:

1. **Control** > **Footswitch Pedal** → selector pill (**TRS X: ...**) → pick the jack until the pill shows **TRS X: UFS**.
2. Long-touch a block → **Learn** → repeat for each switch.

Expression Pedals



Must be calibrated before use (calibration survives a TRS mode change).

1. Plug into a TRS jack.
2. **General** > **TRS** → that jack → **Mode** > **Expression**.
3. Calibrate it, and set **Noise Filter** / **Toe-Heel** / **Response Curve** if needed, from **General** > **Pedals**.

RV Adapters

1. Connect a stereo cable from a TRS jack to the RV's **Relay In** or **CV In**.
2. **General** > **TRS** → that jack → **Mode: Relay Out (via RV)** or **CV Out (via RV)**.
3. Connect a mono / stereo cable from the RV's **Relay Out** / **CV Out** to the amp or effect.

 **Note:** The RV's **9VDC** input is only needed when sending CV.

Connecting MIDI Out

Interface	Connect	Set in Settings → General
5-pin MIDI	Standard 5-pin MIDI cable, MIDI Out port → receiving device	—
TRS Jack	Stereo TRS cable, jack → receiving device	TRS → that jack → Mode: MIDI Out, Type to match
USB	USB-C cable, USB-C port → computer / iOS device (also powers Graviton)	USB → Device
USB Host	USB-A cable, USB-A port → device (Graviton needs its own 9V DC power)	USB → Host
Bluetooth	Wireless, no cable	Bluetooth → pair

 **Note*:** Either the **USB MIDI** or* *USB Host MIDI port can be used at one time, not both.*

Step 2: Programming Buttons



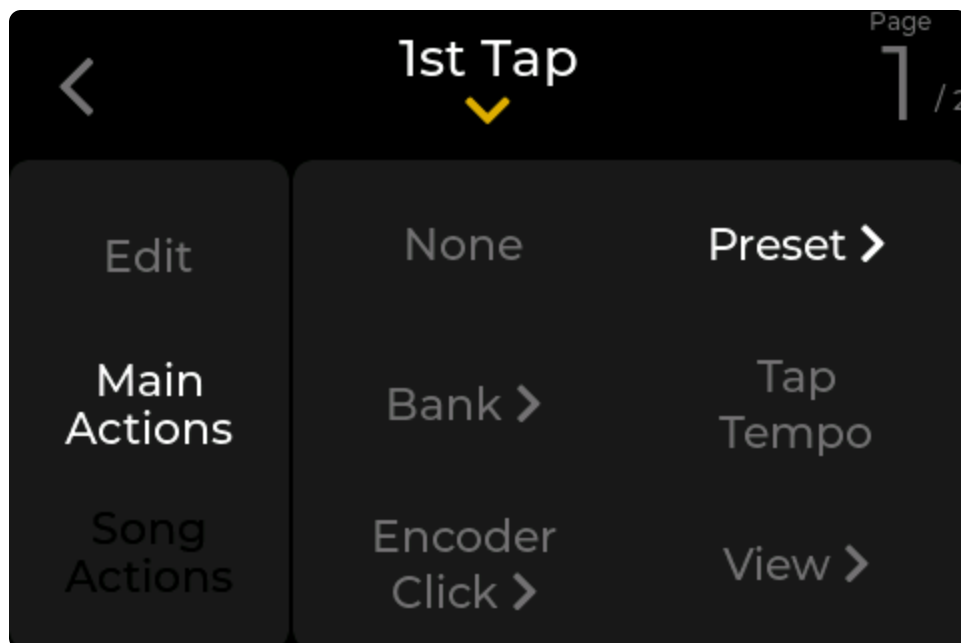
The **Control** views are where you can program buttons.

Most buttons start out as **Bank Preset**: button 1 recalls Preset 1, button 2 recalls Preset 2, and so on — the usual MIDI-controller layout.

That means your footswitches are already mapped 1:1 to the **Preset Library** Presets by default. Just add some PC/CC messages and you have a good starting point.

The next step is adding Actions like **Bank +** or **Bank -**, if you also want to reach Presets in other Banks. Here's how:

Choose Button Actions



- Go to **Control**, then select the controller type — **XY Controller**, **Footswitch Pedal**, or **EC Controller**. Each available button shows as a colorful block.
- If you have more than one controller of the same type, tap the **selector** at the bottom to pick which one — or just press a physical button and the screen switches to it automatically, highlighting the button you pressed.
- **Long-touch** a button block, then pick an Action from **Main Actions** / **Song Actions**.

Commonly Used Actions

- **Bank Preset (1st–10th)**: Recalls the Preset at that slot in the current Bank. Example: **1st** → Preset 1 in Bank 1, Preset 11 in Bank 2, Preset 21 in Bank 3. Default for most buttons.
- **Global Preset (1–120)**: Always the same library Preset, regardless of Bank — e.g. a boost you want in every song, without copying it into all twelve Banks.
- **Bank + / Bank –**: Next / previous Bank.
- **Song Preset scrolling (Song Mode)**: **1st–10th** recalls that slot in the current Song; ← → ↑ ↓ scroll through Song Presets in order (verse → chorus → solo).

Every other Action is listed in **Glossary > Button Actions**.

Multi-Tap (1st Tap, 2nd Tap, 3rd Tap, Hold, Double Tap, Release)

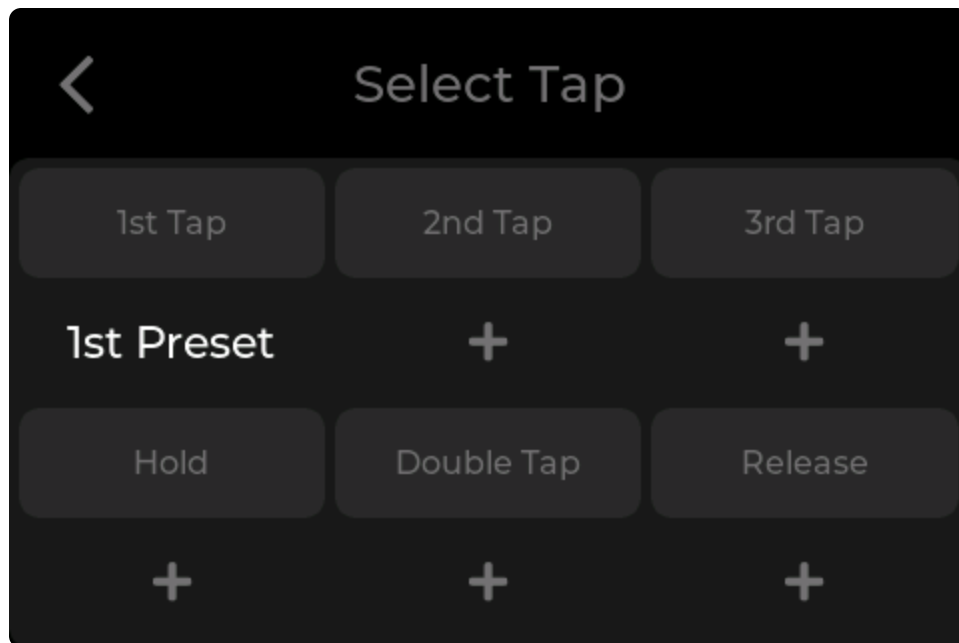
A button can trigger a different action depending on how it's pressed:

- **1st Tap / 2nd Tap / 3rd Tap** — trigger in sequence on repeated presses.
- **Hold** — trigger by pressing and holding.

- **Double Tap** — trigger on a quick double-press.
- **Release** — trigger when the button is let go.

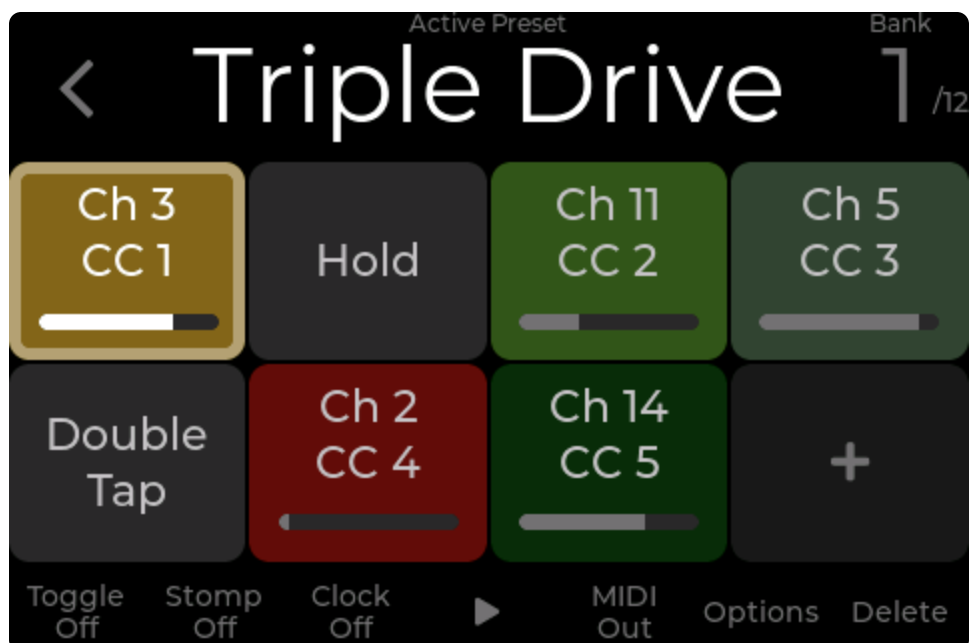
There are two ways to do this:

Global



Tap the **Yellow Down Arrow** to open its Multi-Tap options. This is global — the same behavior applies to that button no matter where you are.

Per-Preset



You can also program Multi-Tap inside a Preset by using the **Tap Marker**, and assign it to a button's 1st Tap. This gives each Preset its own set of Multi-Tap actions, instead of the one global setup used above.

Example: a Preset contains: CC 1 → **Hold** → CC 2, CC 3 → **Double Tap** → CC 4, CC 5

Then, for that Preset's button:

- Normal press sends CC 1.
- Hold sends CC 2 and CC 3.
- Double Tap sends CC 4 and CC 5.

 *Note: Fine-tune timing in Settings → Timing*.*

How this differs from other MIDI controllers

- On most MIDI controllers, each button has a fixed job: button 1 = preset 1, button 2 = preset 2, and so on.
- On Graviton, you decide which button recalls which Preset from the **Preset Library** — in any order you like.
- Because Graviton supports several footswitch pedals, XY, and EC controllers at once, they can all work together as one rig.
- Example: **Footswitch Pedal 1** covers Banks 1–2, **XY** covers Bank 3–4, **EC** covers Bank 5–6. Or they can map to the same Bank if you want.

Step 3: Programming Presets



A Preset is the list of MIDI messages sent when it's recalled. Build and edit Presets in **Home** → **Preset Editor** — see [Home > Preset Editor](#) for the full walkthrough.

Step 4: Programming Expression / EC Controllers

Set what an Expression Pedal sweep or an EC knob turn sends:

- **Expression Pedals** — see [Control > Expression Pedal](#).
- **EC Controllers** — see [Control > EC Controller](#).

Chapter C: Interface Walkthrough

Touch User Interface

There's no separate Edit mode and Play mode — almost every screen lets you program and play at the same time, and changes are saved automatically. There's also no single "right" way, so pick whichever workflow suits you best.

With Most Touch Object

- **Single touch** — select / send (like a left click), or rename.
- **Long touch** — extra options (like a right click): assign, rename, copy, etc
- **Double touch** — shortcut into **Preset Editor** if assigned.

 *Note:** M1 and M2 share the same software, except M2 has more configuration for the extra ports.

Select a View

Tab	What it's for
Home	Build, organize, and play your Presets.
Songs	Build Songs and Setlists, and play through them live.
Control	Program and play with buttons, pedals, and knobs.
Settings	Set up the whole unit — hardware, Song Mode, MIDI routing and filtering, and MIDI monitoring.

Bank Mode / Song Mode

Graviton has two ways to play your Presets. Pick the one that fits the gig — you can switch any time.

Bank Mode

Play Presets straight from the **Preset Library**, one Bank at a time.

- **Bank + / Bank –** or **Bank Jump** — move between Banks.
- **Bank Preset** — recall a Preset in the current Bank.
- **Global Preset** — recall a Preset that stays available in every Bank (e.g. a **Solo Boost**).

Best for quick, hands-on access when you don't need a fixed order.

Song Mode

Play Presets in a set order, organized by Song.

- A **Song** holds up to **10 Presets** from the **Preset Library**.
- A **Setlist** holds up to **10 Songs** from the **Song Library**.
- You can store up to **120 Songs** and **120 Setlists**.

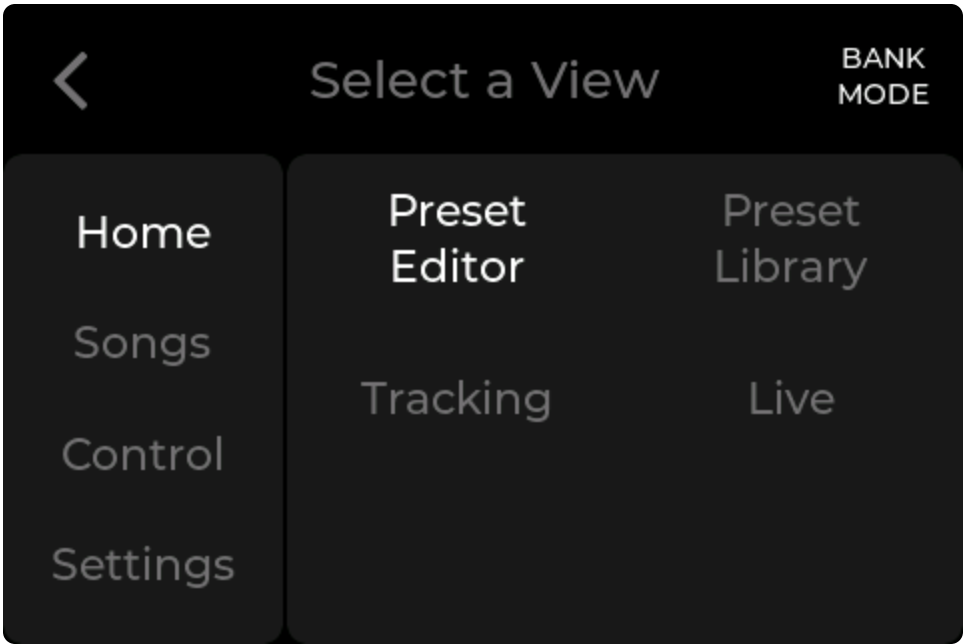
Two ways to play:

- **Setlist Library** → **Setlist** → **Song** — for gigs with a fixed song order.
- **Song Library** → **Song** — pick any Song directly, no Setlist needed. Good for rehearsals or looser sets.

Presets and Songs are shared, not copied: the same Preset can appear in many Songs, and the same Song in many Setlists. Edit one, and every place it's used updates.

 *Note: Controller settings are stored separately for Bank Mode and Song Mode*, so each mode can be set up its own way.*

Home Views



The **Home** tab is where your Presets live. It has four views: two for building and storing Presets, and two for watching and playing them.

View	Shows	Use it to...
Preset Editor	The MIDI messages in the current Preset	Build and edit a Preset
Preset Library	All 120 Presets (12 Banks × 10)	Send, copy, swap, and organize Presets
Tracking	The current PC and CC values of your gear	See where your gear is set, and adjust it
Live	The current Preset name, extra large	Play on stage without accidental touches

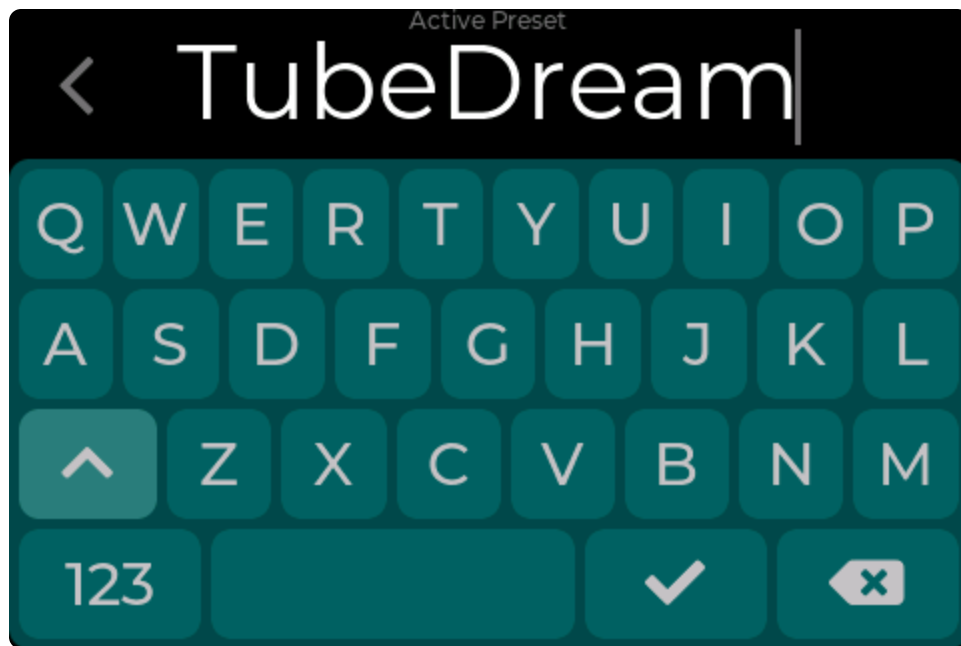
 *Note: The Preset Library is the one shared pool of Presets. Every controller and every Song recalls Presets from here, so an edit in Preset Editor shows up everywhere that Preset is used.*

Home > Preset Editor

Quick Start

Programming a preset is very simple:

- Go to **Home** → **Preset Editor**.
- Tap “+” to add a **Message** (up to **16** per Preset).
- When the Preset is triggered by an Action, all messages are sent **in sequence**.



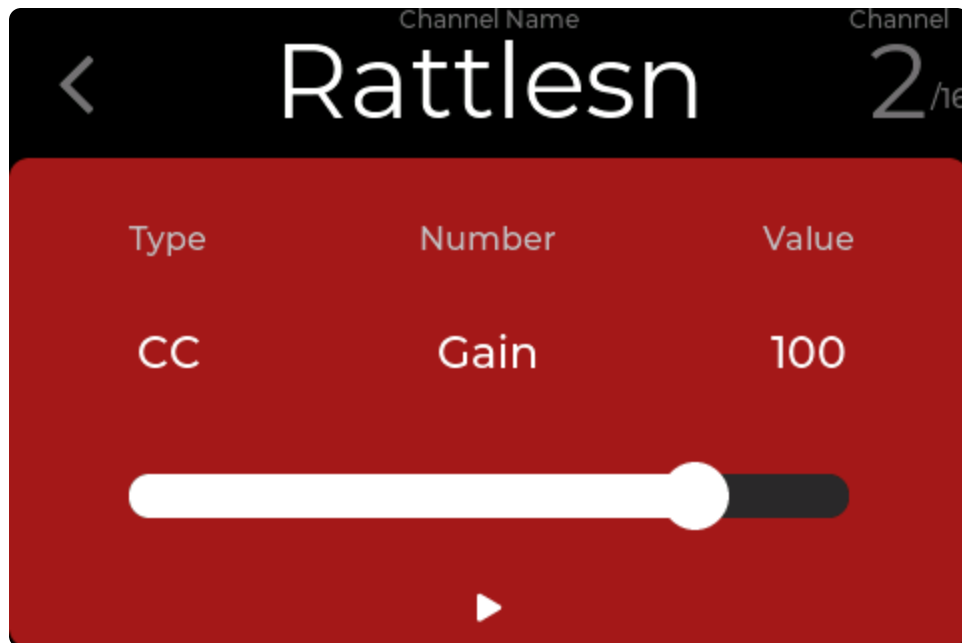
From the top:

- **Preset Name:** Shows the **Current Preset** name. Tap to rename.
- **Clock LED / BPM:** Blinks in sync with the BPM when Clock is running.
- **Bank:** Shows current **Bank**. Tap to switch to another **Bank**.

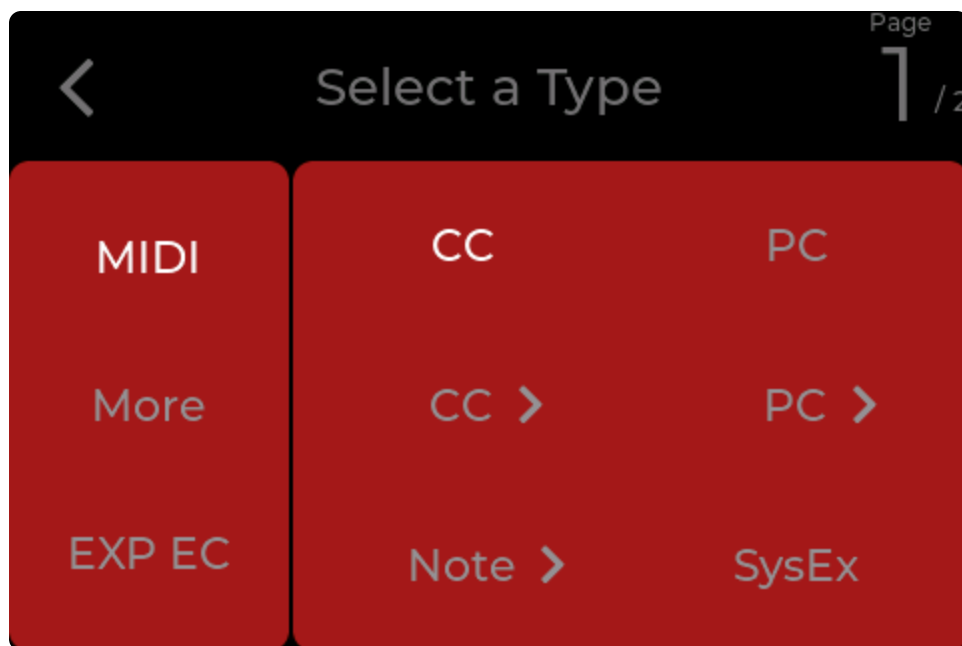
Messages grid:

- **Message Blocks:** Each block displays a **Message** and its parameters.
- **Single Touch:** Edit selected **Message**.
- **Long Touch:** Move / Swap / Insert / Disable / Delete / Send the selected **Message**.

Opening a message for editing:

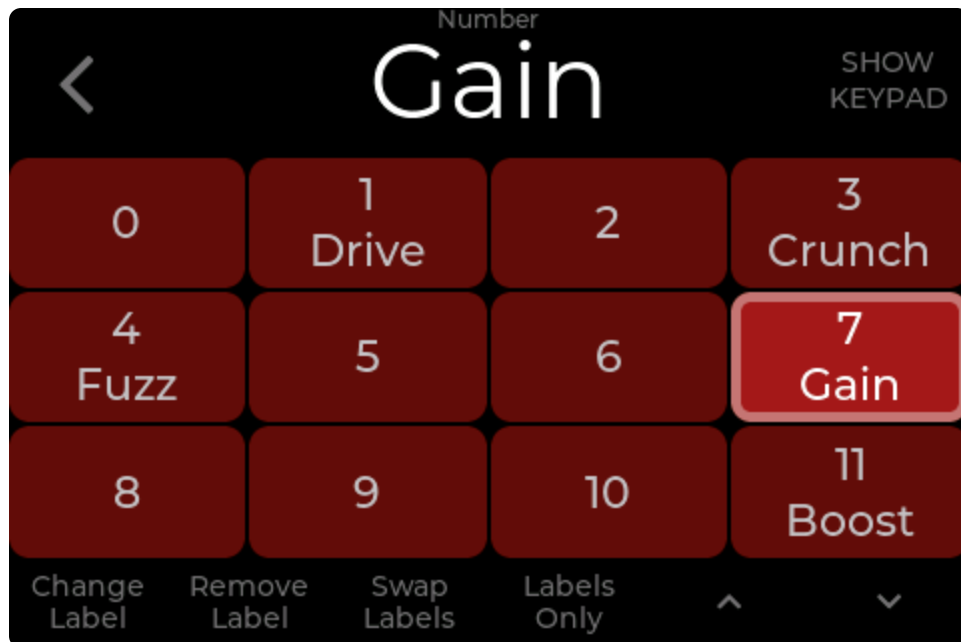


- **Type:** Tap to change the **Message** type.

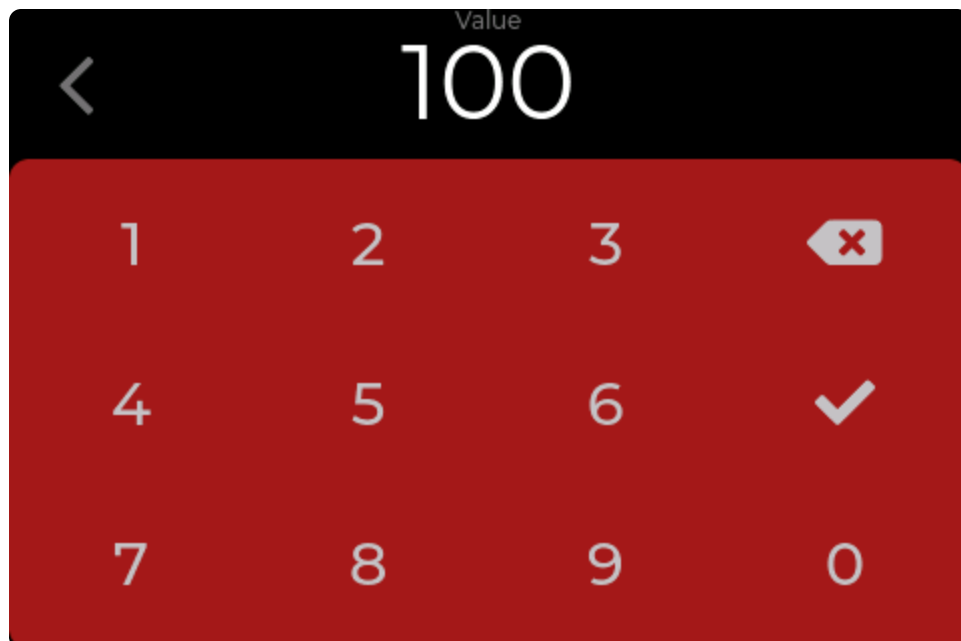


- **Number / Note / BPM / Name** (depending on the selected **Type**) : Tap to input a number by keypad.
 - If the type is **CC/PC/Counter CC/PC or Note**, you can switch to the “**List**” picker (top right corner), where you can view the list with attached **Labels** (Up to 255).
 - **Add / Change / Remove Label**
 - **Swap Labels** — copy/swap *all* labels from the current MIDI channel to another channel (**Swap all Labels: Channel N > Channel:** → confirm)

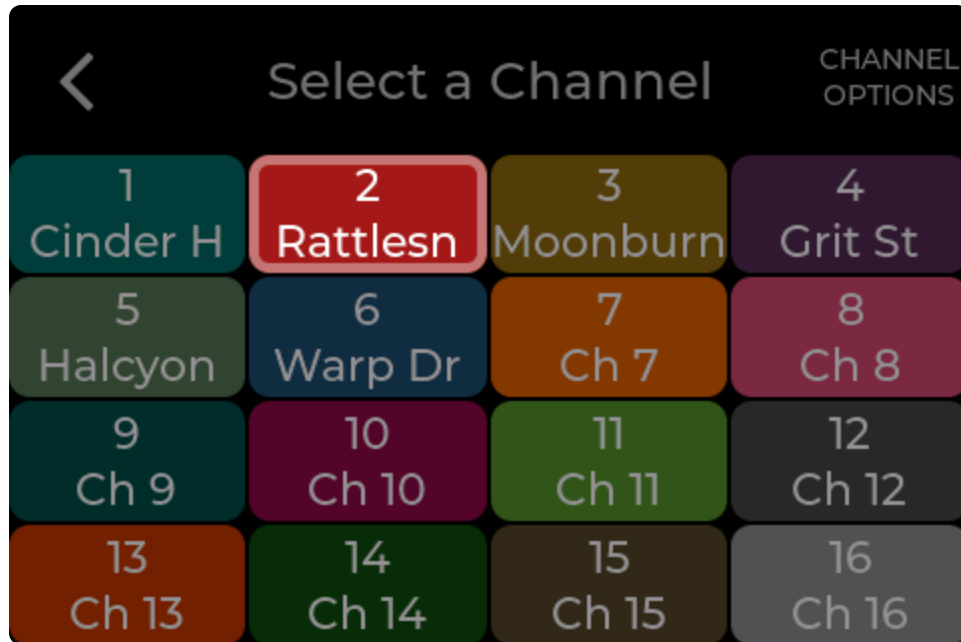
- **Labels Only** — show only numbered entries that already have names



- **Value / Velocity:** Tap to input a value number by keypad.
 - **Slider:** If the Type is **CC**, **Counter CC**, **CV** or **Pitch Bend** a slider control will be shown, allowing you to drag and change the value by touch.
 - If the values are **Counter CC/PC** or **CC/Clock Expression** values, their colors will be yellow (changing values) instead of white (fixed values).



- **Channel:** Tap to change to another **MIDI Channel** (1 to 16). In the same screen, tap **CHANNEL OPTIONS** in the title bar for **Rename** or **Change Color** (when channel colors are enabled in **Editor Options**).
- **Channel Name:** Tap to rename the current **MIDI Channel**.



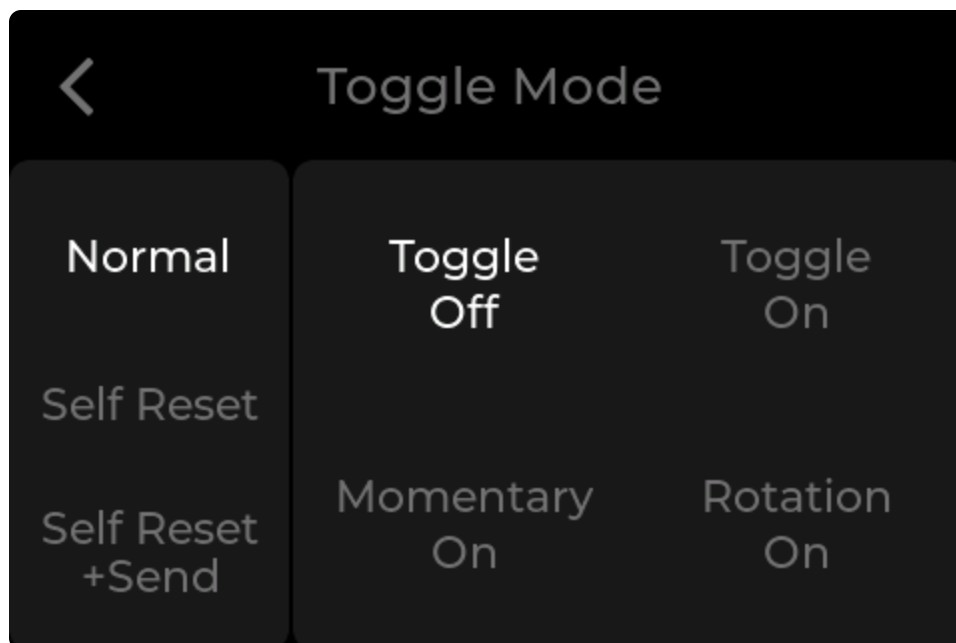
- ► : Send this **Message**.

Bottom menu bar:

- **Toggle Mode** — per-Preset **Toggle** / **Momentary** / **Rotation** status. See **Toggle Mode** below.
- **Stomp On/Off** toggle for this Preset (sends without becoming the main Current Preset). See **Stomp Mode**.
- **Clock** — Global BPM / clock status. Tap for the BPM keypad. See **Clock** below.
- ► — send / play this **Preset**.
- **MIDI Out** — per-Preset port mask. See **MIDI Out (per Preset)** below.
- **Options** — opens **Editor Options**:
 - **Layout** — **Grid** | **List**
 - **Value Sliders** — **On** / **Off** (change Values with touch instead of keypad)
 - **Monochrome** — **On** / **Off** (On = grayscale message backgrounds instead of channel colors)
- **Delete** — remove the last message.

⚠ Note: When there are more messages than fit on one screen, the pager buttons (◀ / ▶) will appear.

Toggle Mode



Toggle Mode changes what a Preset does on repeated presses. The popup has three tabs: **Normal**, **Self Reset**, and **Self Reset + Send**.

Normal

Choice	What each press does	Good for
Toggle Off (<i>default</i>)	Sends the Preset the same way every time	Scenes, patch changes
Toggle On	Alternates between On and Off	Delay on/off, boost in/out
Momentary On	Sends On when pressed, Off when released	Hold-to-use effects, talkback, mute
Rotation On	Sends the next step (A → B → C → A...)	Three delay times, three amp channels

What gets sent as **Off** — two ways:

- **Auto Off** — program only the **On** messages. Graviton works out the **Off** messages for you:
 - **CC** → same CC with value **0**
 - **Note** → same Note with velocity **0**
 - **Switch / Relay** → the opposite state (Open ↔ Close)
- **A/B** — insert an **A/B** message as a divider. Messages **before** it are sent as **On**, messages **after** it as **Off**. Use this when On and Off need completely different messages.

Self Reset

When you move to a **different** Preset, this Preset quietly goes back to a starting state (nothing is sent):

- **(Toggle On) Self Reset To On / To Off** — returns to **On** or **Off**.

- **(Rotation On) Self Reset To A** — returns to the first step.

Self Reset + Send

Same as **Self Reset**, but the starting state is also **sent** — handy for keeping an external multi-effects unit in sync:

- **(Toggle On) Self Reset To On + Send / To Off + Send**
- **(Rotation On) Self Reset To A + Send**

Stomp On/Off

A Preset with **Stomp On** (shown with a **lightning** mark) still sends all of its messages, but it does **not** become the current Preset. Use it for background actions — delay on/off, a boost, tap tempo — without leaving the Preset you're playing.

Example: you're on "Verse Clean". A **Stomp On** Preset on another button turns the delay on. **Live** still shows "Verse Clean" — only the delay changed.

⚠ *Tip: Combine Stomp On with Toggle On so the same button turns the effect on and off, without changing the current Preset.*

Clock



Send tempo from Graviton to a looper, delay, or drum machine. Set one **global BPM** here and choose how it's sent:

Type	Sends tempo as...	Where
MIDI Clock	MIDI Clock over 5-pin, TRS MIDI, USB, Bluetooth	Choose ports in MIDI Routing → Clock
Analog Clock	Tap pulses for devices with a tap-tempo input	TRS Tip/Ring (M2 also Relay Tip/Ring). Set the TRS jack to Switch Out first in Settings → TRS

Set the BPM — in **Home** → **Preset Editor**, tap the clock button on the bottom bar to open the BPM keypad:

- Enter **40–300** to set the BPM (this also turns **MIDI Clock** on).
- Enter **0** to turn all clocks off.
- Use **.00 / .25 / .50 / .75** on the bottom row for quarter-BPM steps.

Choose outputs — from the BPM keypad, open **Clock Mode**:

- **MIDI Clock** — turn MIDI Clock on or off.
- **Analog Clock** → — pick which outputs send Analog Clock (M1: TRS 1–2 Tip/Ring; M2: TRS 1–4 Tip/Ring + Relay Tip/Ring).

⚠ *Tip: To give a Preset its own BPM, add a MIDI Clock or Analog Clock message to it. Recalling that Preset overrides the global BPM.*

More options in **Settings** → **Clock**:

- **While Switching Presets** — **Keep All Clocks Running** or **Stop All Clocks**.
- **Analog Tap Clock Cycle** — **4 Taps** or **Continuously**.

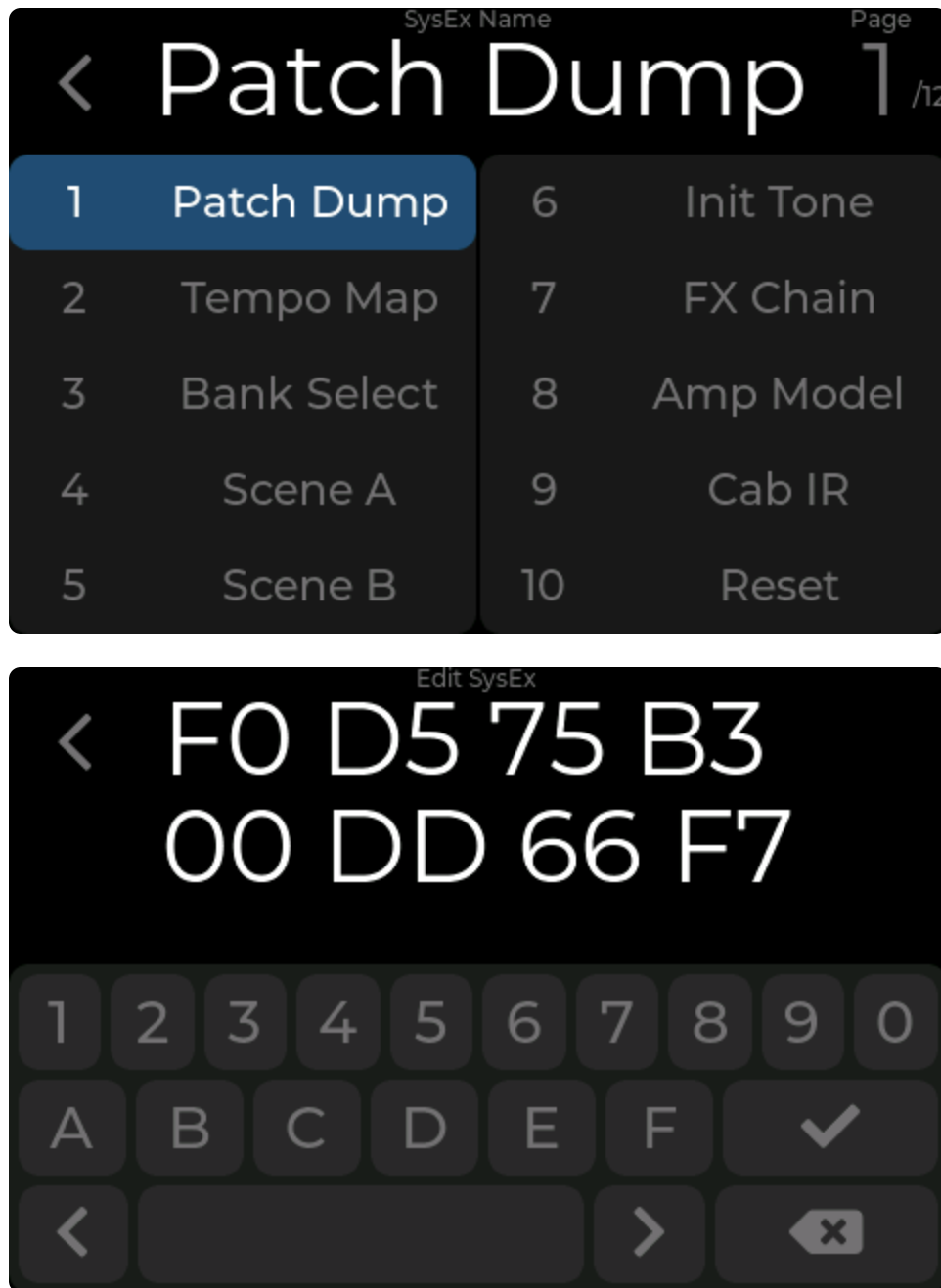
MIDI Out (per Preset)

Choose which ports *this* Preset is sent to, instead of all of them.

Example: your delay is on TRS 1 and your amp is on USB. For a delay Preset, leave only **TRS 1** on. For an amp Preset, leave only **USB** on.

⚠ *Tip: This is different from Channel Filter, which limits MIDI channels (1–16). To turn a port off for all Presets, use MIDI Routing → Out.*

SysEx Library



Stores up to **120 SysEx messages** across **12 Pages**. To open it, add a **SysEx** message in **Preset Editor** and tap its Message / Value field.

From the top:

- **SysEx Name** — the selected SysEx message. Tap to rename.
- **Page** — the current Page. Tap to switch Pages.
- **SysEx 1–120:**
- **Single touch** — pick this SysEx message for the Preset you're editing.
- **Double touch** — edit the message (hexadecimal, up to **64 bytes**).

- **Long touch** — **Edit** / **Rename** / **Copy** / **Swap** / **Delete**.

 *Note: For messages longer than 64 bytes, split them into several SysEx entries — the first starting with F0, the last ending with F7 — and add them to the Preset in order.*

Relay and Control Voltage

Use **Relay** when the amp or pedal only has a *footswitch jack*, not MIDI — channel switch, FX loop, analog boost. **M2** has a Relay port; **RV Adapters** add more on M1 or M2.

Example: Preset “Lead” sends PC to the modeler *and* a Relay **Close** on Tip so the real amp jumps to lead channel.

Use **CV** when a synth / modular wants voltage, not MIDI notes.

Relay Port

- A Relay Port is available on the Graviton M2, and more ports can be added to any M1 or M2 with **RV Adapters**.
- **Relay Ports** function as **Virtual Footswitch Pedals**, which can send **Switching messages** to devices with **Footswitch Inputs**, such as amps or effects, for changing amp channels / switching effects on and off, and more.
- Each **Relay Port** has a **Tip** and **Ring** output, which can control two Footswitch Inputs separately.
- For single device use, a mono cable can be used (**Tip** only). You can also use a stereo to dual mono splitter cable for **Tip** and **Ring** outputs, allowing you to control two devices at the same time.
- Add **Switch** messages in **Preset Editor**: choose **Jack** (TRS / Relay), **Pin** (**Tip** / **Ring**), and **State** (**Close** / **Open**, etc.).
- If **Toggle** or **Momentary** is enabled, the switched state during the 'Off' state will be the reverse of the 'On' state.

Control Voltage Port

- A **CV Port (CV)** is available on the Graviton M2, with more can be added to any M1 or M2 with **RV Adapters**.
- **CV Ports** can send **CV messages** to devices with **CV interfaces**, such as synthesizers and pedals. They can also function as **Virtual Expression Pedals**, allowing control of devices with expression pedal input jacks.
- Each **CV Out** has a **Tip** and **Ring** output, each controllable separately
- For single device use, a mono cable can be used (**Tip** only). You can also use a stereo to dual mono splitter cable for **Tip** and **Ring** outputs, allowing you to control two devices at the same time.
- Start using CV by adding CV **messages** to the **Preset Editor**, with the **Jack**, **Pin** / **Range** and **Output** (%) you want (static output).

- You can also control the **CV messages Output %** dynamically with external **Expression Pedals** connected to the M1/M2 **TRS Jacks** or with the **EC Controllers**, in the same way with **CC messages**.
- To drive the Expression Input (as a virtual Expression Pedal) of another pedal, ensure that the **Range** matches the maximum voltage allowed for that pedal (0-3.3V, 0-5V, or 0-9V).

Home > Preset Library



The Preset Library holds all **120 Presets** (12 Banks × 10). It's one shared pool: every controller (XY, EC, footswitch pedals, **MIDI In**) recalls Presets from here, in both **Bank Mode** and **Song Mode**.

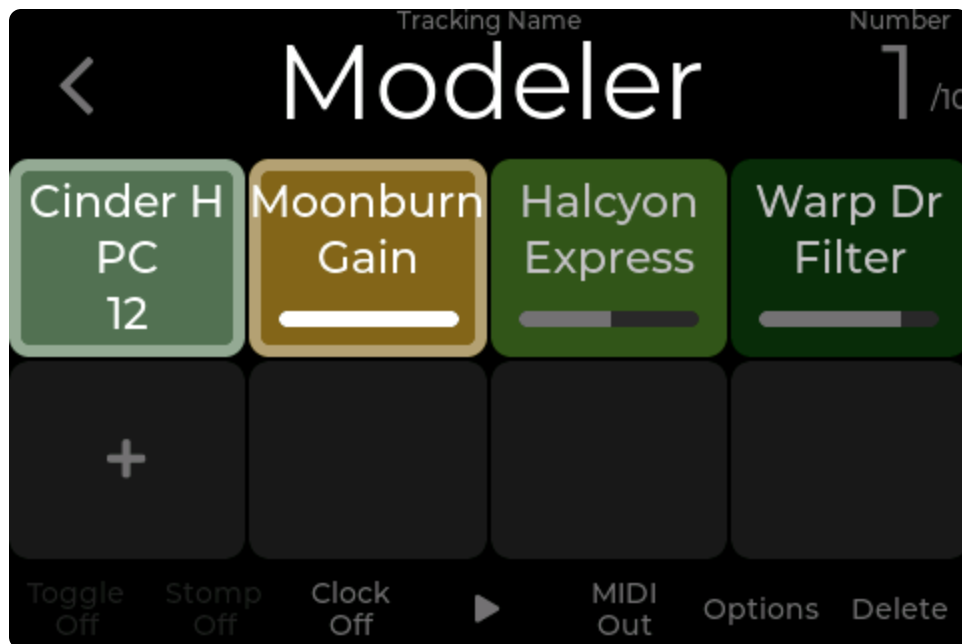
On screen, the current Preset is highlighted, each Preset shows its own color, and **Stomp On** Presets show a lightning mark.

From top to bottom:

- **Preset Name** — the current Preset. Tap to rename.
- **Bank** — the current Bank. Tap to switch Banks.
- **Preset blocks** — Preset numbers and names:
- **Single touch** — send the Preset.
- **Double touch** — open it in **Home** → **Preset Editor**.
- **Long touch** — **Edit Preset** / **Rename Preset** / **Copy Preset** / **Swap Preset** / **Reset Preset** / **Change Color**.
- **Toggle LED** — shows **On** / **Off** for **Toggle On** and **Momentary On** Presets, or the current step for **Rotation On**.
- **Layout** (bottom bar) — switch between **Grid** and **List**.

Example: to start Bank 2 with the same clean tone as Bank 1, long-touch Preset 1 → **Copy Preset** → Preset 11, then adjust only what's different.

Home > Tracking



Tracking shows the **real, current** state of your other MIDI gear on Graviton's screen — patch number, effect on/off, delay mix — no matter whether the change came from Graviton or from the device itself.

Example: you turn the delay mix knob on your modeler. Tracking updates CC 11 on screen, so you always know where it's set.

⚠ Note: For changes made on the device to show up, the device has to send MIDI back to Graviton. USB and Bluetooth MIDI work both ways on one connection. 5-pin and TRS MIDI only go one way, so you need a second cable from the device's MIDI Out to Graviton's MIDI In.

How it works

Open **Select a View** → **Home** → **Tracking**.

1. **Set up a Tracking Preset.** Add the PC and CC messages you want to watch — just like programming a normal Preset. You can have up to **10 Tracking Presets**; tap the number at the top right (e.g. **1 / 10**) to switch between them.

2. **Graviton watches all MIDI traffic in both directions:** - **Outgoing** — anything Graviton sends, from any source (an XY or FxW button, an EC knob, a Preset, ...). - **Incoming** — anything the device sends back, whether you pressed something on the device or it sends automatic status updates (often called **MIDI Feedback** or **Status Updates** in the device's manual).
3. **When a message matches, the screen updates:** - **PC** — same channel → the PC **number** is shown. - **CC** — same channel **and** CC number → the CC **value** is shown.

Each PC and CC in a Tracking Preset is also a **control**: touch it to change the value and send it straight to the device.

Example

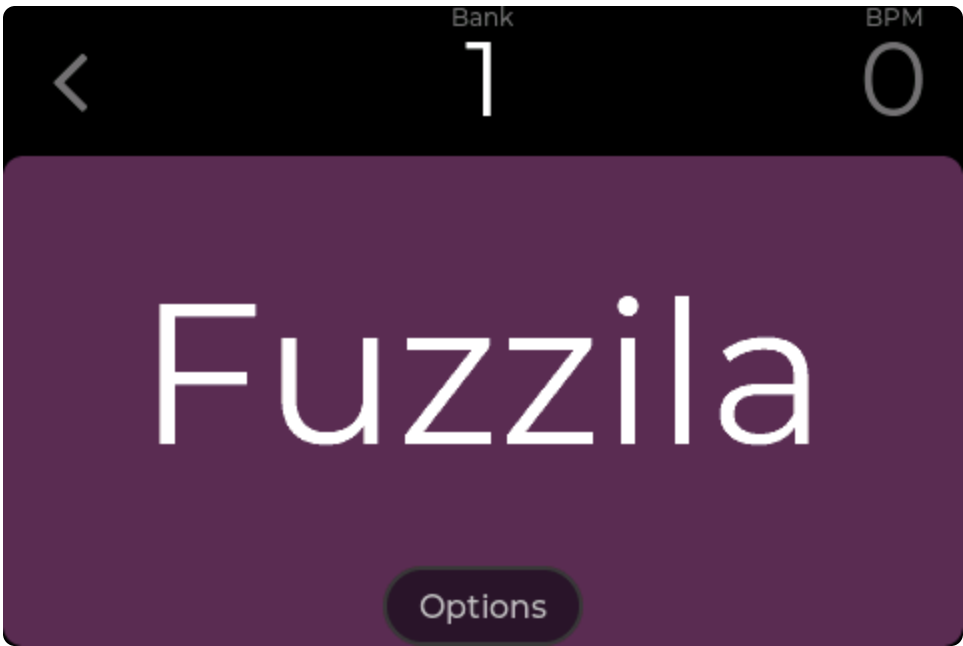
Your device changes patch on **PC**, turns **FX1 / FX2 / FX3** on and off with **CC 1 / 2 / 3** (value 127 = on, 0 = off), and sends updates back when you change something on it.

To watch the patch number and all three effects, add these to a Tracking Preset (any value):

- **PC** — patch
- **CC 1** — FX1
- **CC 2** — FX2
- **CC 3** — FX3

What you'll see:

You do this	MIDI sent	Tracking shows
Switch to Patch 1 from Graviton	Graviton → device: PC 1	PC 1
Switch to Patch 2 on the device	Device → Graviton: PC 2	PC 2
Turn FX1 on from Graviton	Graviton → device: CC 1 = 127	CC 1 = 127 (on)
Turn FX1 off on the device	Device → Graviton: CC 1 = 0	CC 1 = 0 (off)



A stage view: the current Preset name is shown extra large so you can read it from a few feet away.

To avoid accidental changes while you play, every touch control is turned off except **Back** (<) and **Options** (⋮).

- **Toggle On / Momentary On** Presets show an **On / Off** LED.
- **Rotation On** Presets show the current step (**A, B, C, ...**).

Live Options

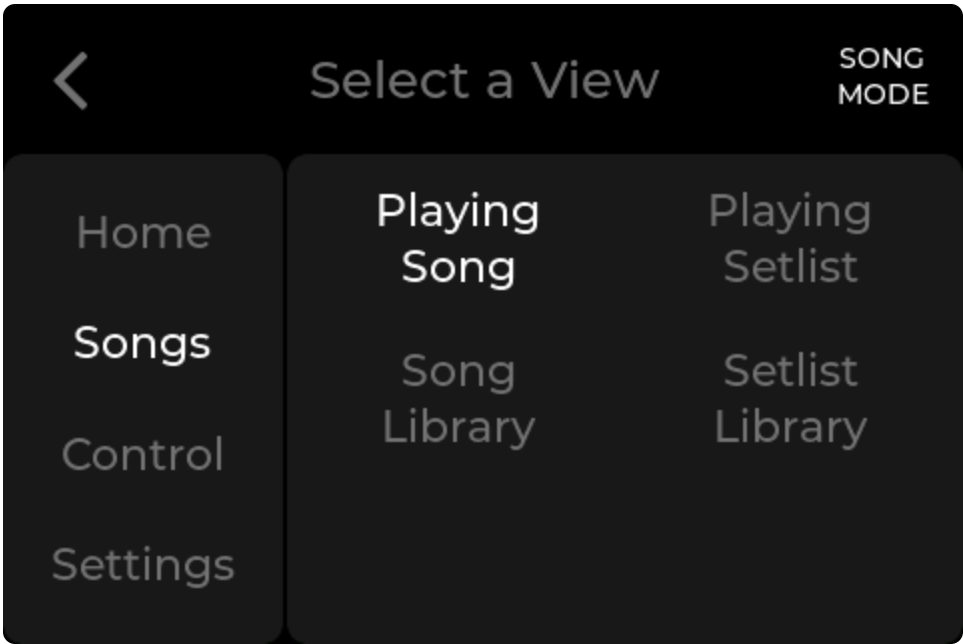
Tap **Options** (⋮) to choose what each area of the screen shows:

Area	Choices
Center	Preset, Bank, BPM, Stomp Preset (+ Song, Setlist in Song Mode)
Top Center	Same as Center
Top Right	Same as Center
EC/EXP Slider	While Active (only while the EC knob or Expression Pedal moves) or Always On

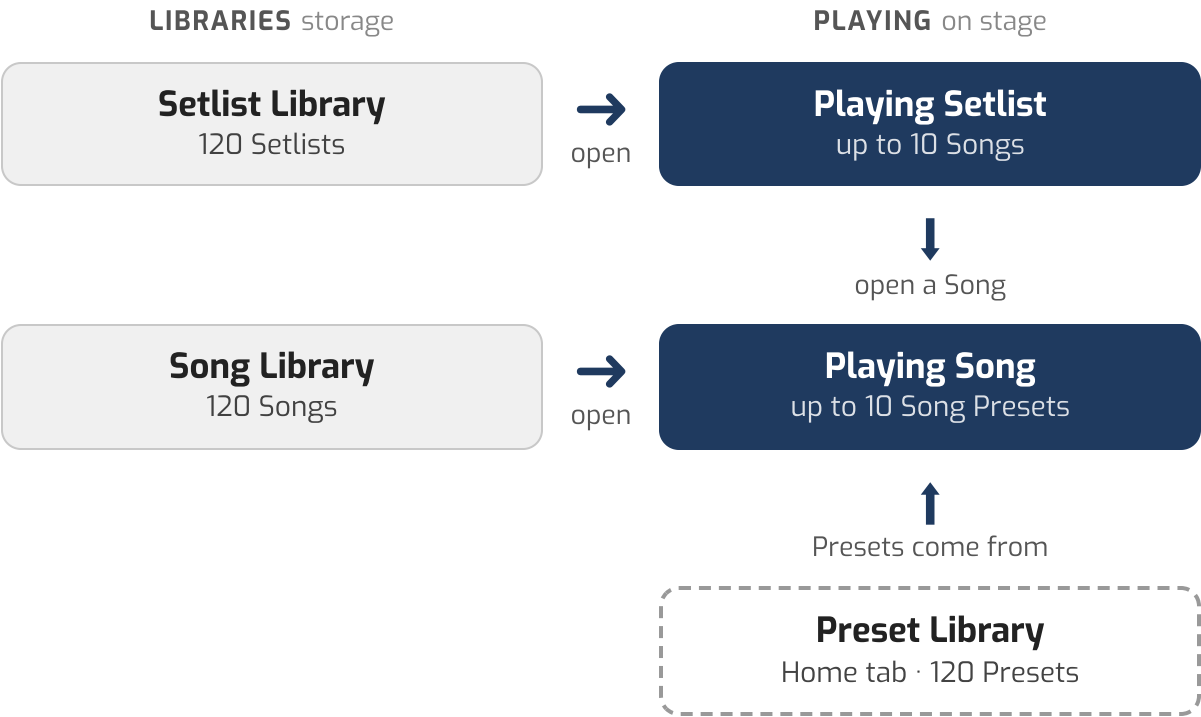
Tap **RESET** to go back to the defaults. **Bank Mode** and **Song Mode** each keep their own Live Options.

Example: in **Song Mode**, show **Preset** in the center, **Song** at the top, and **BPM** at the top right.

Songs Views



The **Songs** tab has four views: two **libraries** where everything is stored, and two **playing** views for what you're using right now.



View	Holds	Single touch opens	Use it to...
Setlist Library	Up to 120 Setlists	Playing Setlist	Store and organize your Setlists
Playing Setlist	Up to 10 Songs, in order	Playing Song	Pick the next Song during the set
Song Library	Up to 120 Songs	Playing Song	Store Songs, or play one without a Setlist
Playing Song	Up to 10 Song Presets	— (sends the Preset)	Play the Song's Presets

Two ways into **Playing Song**:

- **With a Setlist:** Setlist Library → Playing Setlist → Playing Song — for gigs with a fixed order.
- **Without a Setlist:** Song Library → Playing Song — for rehearsals or looser sets.

 *Note: Nothing is copied. A Setlist points to Songs in the Song Library, and a Song points to Presets in the Preset Library — edit one and every Setlist or Song using it updates.*

 *Tip: Need more than 10 Songs in a set? Split it across Setlists stored next to each other in the Setlist Library — e.g. Friday 1/3, Friday 2/3, Friday 3/3 — then set Settings → Song Mode → At Setlist Ends to Next Setlist. When the last Song of 1/3 ends, Graviton moves straight on to 2/3.*

Songs > Playing Song



The main view while performing in **Song Mode**: the current Song's Presets, in the order you'll use them (intro → verse → chorus → solo).

From the top:

- **Playing Song** — the Song's name. Tap to rename.
- **Song Presets 1–10:**
- **Single touch** — send the Song Preset.
- **Double touch** — open it in **Preset Editor**.
- **Long touch** — **Edit Preset** / **Rename Preset** / **Change Preset** / **Swap Preset** / **Remove Preset**.
On an empty slot, this adds a Preset.
- **Toggle LED** — shows **On** / **Off** for **Toggle On** and **Momentary On** Presets, or the current step for **Rotation On**.
- **Stomp On** Presets show their On / Off state, while the Song Preset you're on stays highlighted — stomping a delay on top doesn't move the highlight.

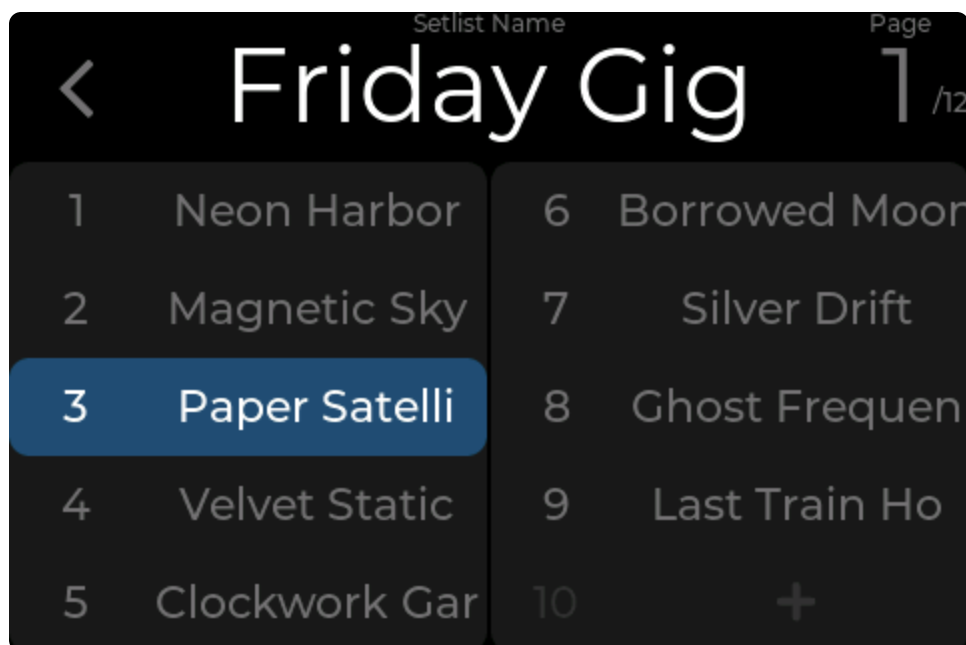
Song Options

Settings for **this Song only**:

- **Song BPM** — the tempo set when the Song loads. **Off** stops the clock instead of keeping the previous Song's tempo. Quarter-BPM steps are supported.
- **Startup Preset** — a Preset sent automatically when the Song loads:
- **None**
- **Song Preset** — **1st–10th** from this Song.
- **Global Preset** — any of the 120 Presets in the **Preset Library**.

 *Note: Settings that apply to all Songs — At Song Ends, At Setlist Ends, Presets Self Reset on Song Change, and Layout — are in Settings → Song Mode.*

Songs > Playing Setlist



The Setlist you're playing tonight, in running order. Use it to pick the next Song.

From the top:

- **Playing Setlist** — the Setlist's name. Tap to rename.
- **Songs 1–10:**
- **Single touch** — open and play the Song in **Playing Song**.
- **Long touch** — **Open Song** / **Rename Song** / **Change Song** / **Swap Song** / **Remove Song**.

Example: the encore isn't on the Setlist. Long-touch the last slot → **Change Song** and pick it from the **Song Library**.

Performance tips

- Assign **Song Library** ±, **Setlist Library** ±, **Song Jump**, or **Navigate** Actions to footswitches to move through the set hands-free.
- Use **View Jump** / **View Toggle** to get to **Playing Setlist** or **Playing Song** quickly.
- What happens at the end of a Song or Setlist is set in **Settings** → **Song Mode** (**At Song Ends**, **At Setlist Ends**).

Songs > Song Library



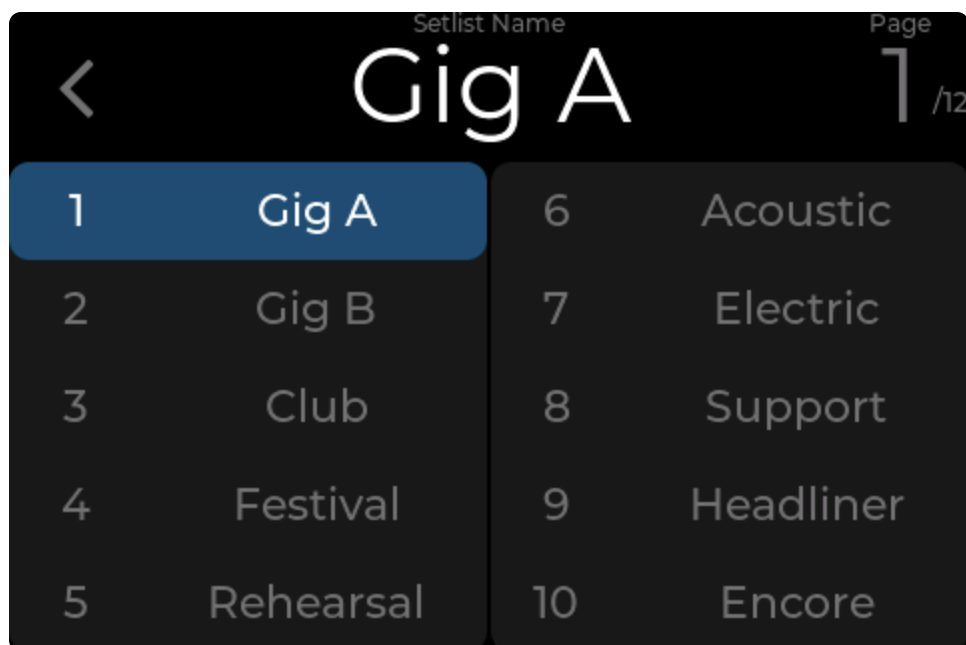
Where all your Songs are stored, across **12 Pages**.

From the top:

- **Song Name** — the selected Song. Tap to rename.
- **Page** — the current Page. Tap to switch Pages.
- **Songs 1–120:**
- **Single touch** — open and play the Song in **Playing Song**.
- **Long touch** — **Open Song** / **Rename Song** / **Copy Song** / **Swap Song** / **Delete Song**.

 *Note: To add a Song to a Setlist, open the Setlist in Playing Setlist and long-touch a slot.*

Songs > Setlist Library

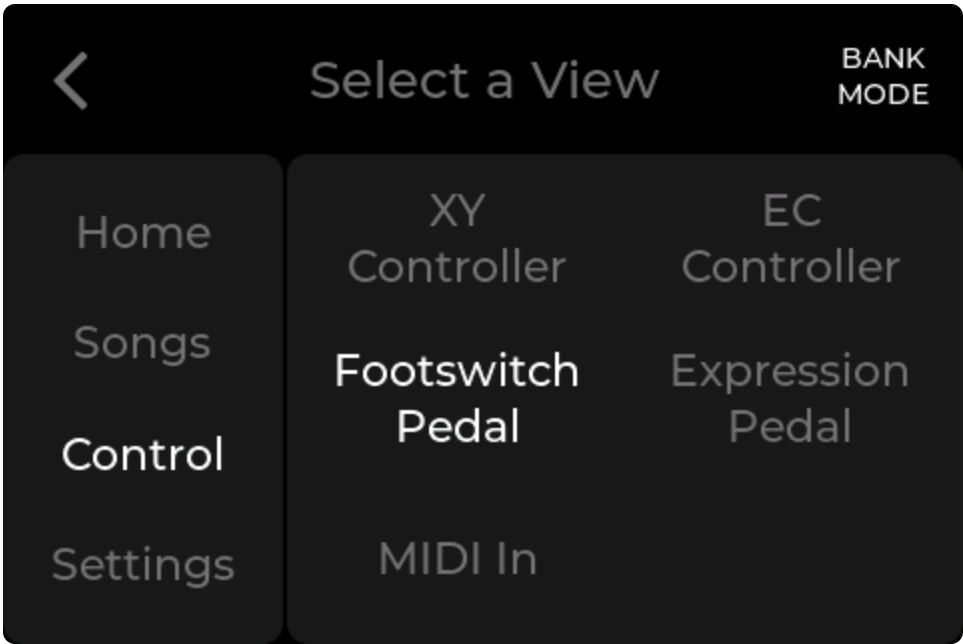


Where all your Setlists are stored, across **12 Pages** — e.g. one for a club gig, one for a festival, one for an acoustic set.

From the top:

- **Setlist Name** — the selected Setlist. Tap to rename.
- **Page** — the current Page. Tap to switch Pages.
- **Setlists 1–120:**
- **Single touch** — open the Setlist in **Playing Setlist**.
- **Long touch** — **Open Setlist** / **Rename Setlist** / **Copy Setlist** / **Swap Setlist** / **Delete Setlist**.

Control Views



The **Control** tab is where you program and play each connected controller. Every control is one of three types, and each type sends something different:

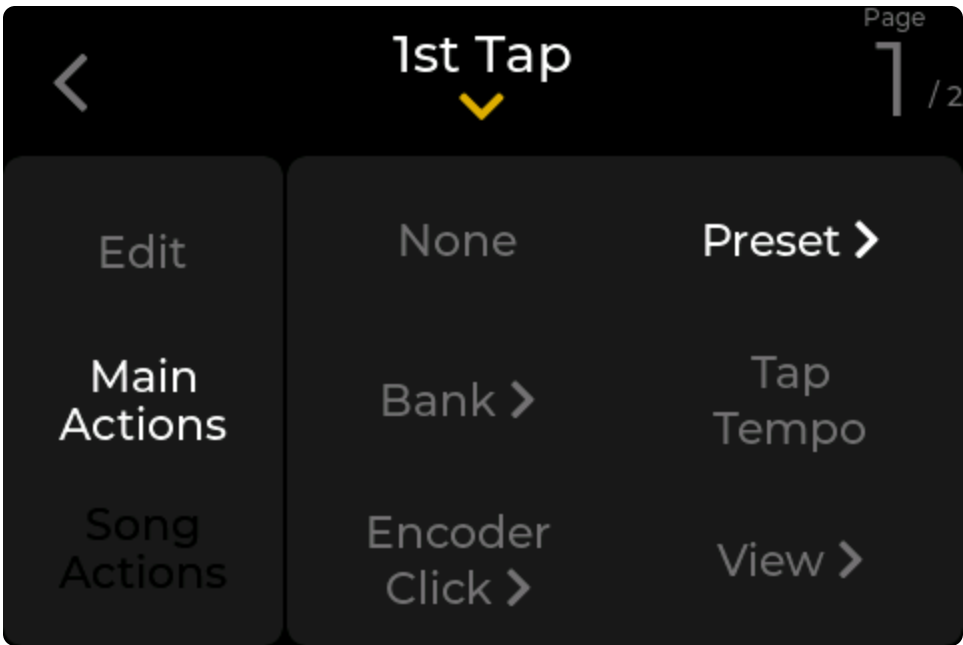
View	Buttons → Actions	Knobs → Encoder messages	Pedals → Expression messages
XY Controller	✓		
Footswitch Pedal	✓		
MIDI In	✓ (triggered by incoming MIDI, not a press)		
EC Controller	✓ (push a knob)	✓	
Expression Pedal			✓

- **Actions** — Presets, Bank ±, Tap Tempo, ... (see [Step 2: Programming Buttons](#)).
- **Encoder messages** — sent as you turn a knob. Pushing a knob works like a button and sends an Action.
- **Expression messages** — sent as you sweep a pedal.

 *Note: Buttons work the same in every view — the same Actions, Multi-Tap, and settings. See [Control > Button Settings](#) below; learn them once, use them everywhere.*

Control > Button Settings

Per Button



Long-touch a button block (or single-touch an empty block) to open this menu:

Tab	What it does
Edit	Jump to Preset Editor for whatever Preset this tap recalls (gray if the tap isn't a Preset).
Main Actions	Pick what this tap <i>does</i> — recall a scene, step banks, open a screen, and so on.
Song Actions	Pick what this tap <i>does</i> — Song Presets, setlists, song jumps. Only available in Song Mode
Focus	Choose whether pressing this controller moves the display to that Control view, stays put, or needs a second press.
Navigate	Assign “menu cursor” moves for Playing Song / Playing Setlist (scroll and confirm without a switch per line).
Learn / Trigger	Learn teaches an unknown Universal footswitch step. Trigger (MIDI In only) sets which incoming message fires this virtual button.

Per Controller



Setting	What it does
Global Bank	Shares one bank with other controllers set to On , or keeps a separate bank when Off .
Bank Min / Bank Max	Sets the range of Banks this controller can reach.
Bank Wrap	Decides whether going past Bank Max / Bank Min jumps to the other end or stops.
Focus	Decides whether the screen jumps to this controller's view when you press one of its buttons.
Self Reset Tap	Decides whether a button waiting on its 2nd / 3rd Tap goes back to 1st Tap when you press another button.

Global Bank

- **On** — this controller shares one bank with every other controller set to **On**. Changing the bank on any of them (**Bank +**, **Bank -**, **Bank Jump**) changes it on all of them.
- **Off** — this controller has its own bank. Changing it doesn't affect any other controller.

Example: set both footswitch pedals to **Off** so pedal 1 stays on Banks 1–6 while pedal 2 stays on Banks 7–12.

Bank Min / Bank Max

Set the range of Banks this controller can reach — **Bank +**, **Bank -**, and **Bank Jump** all stay inside it.

Bank Wrap

This decides what happens when you go past the end of the range:

- **On** — **Bank +** on **Bank Max** jumps to **Bank Min**, and **Bank -** on **Bank Min** jumps to **Bank Max**.
- **Off** — the bank stops at **Bank Min** / **Bank Max**.

Example: **Bank Min** 1, **Bank Max** 4. On Bank 4, **Bank +** goes to Bank 1 with **Bank Wrap On**, and stays on Bank 4 with it **Off**.

Focus

Focus decides whether the screen jumps to a controller's view when you press one of its buttons.

Choice	Does the screen jump to this controller?	Does the Action fire?
Same View (default)	Only if you're already on that type of Control view (ex: F10W > M2, 1st XY > 2nd XY)	Yes
Press Once	Yes, on the first press	Not on the first press; yes after that
Always	Yes, on every press	Yes
Off	No — the screen stays where you are	Yes

One setting for the whole controller, or one per button

- From the controller's bottom menu, **Focus**:
- **All buttons** — pick one choice for every button on this controller (the M2's built-in switches, one F10W, one XY, ...).
- **Per button** → **On** — each button uses its own Focus setting.
- From a button's Action popup (long-touch a block), **Focus**:
- **This button** — pick a choice for just this button.
- **All buttons** → **On** — this button follows the controller-wide setting again.

Self Reset Tap

Say a button is waiting on its **2nd Tap** or **3rd Tap**, and you press a different button. **Self Reset Tap** decides what happens to the first button:

Choice	The first button...
Off (default)	Stays where it was — its next press is still the 2nd / 3rd Tap
On	Goes back to 1st Tap , silently
On + Send	Goes back to 1st Tap and sends its 1st Tap Action

Example: button 1 steps through three delay times (**1st / 2nd / 3rd Tap**). With **On**, after pressing another button, button 1 starts over at the first delay time on its next press. With **On + Send**, the first delay time is sent right away.

Set separately for each controller (the popup title is **Self Reset 2/3rd Tap**).

Control > XY Controller



The buttons of your paired **XY Controllers** — up to **10**, each with its own view.

From the bottom menu:

- **Controller** — switch between paired XY Controllers.
- **Layout** — **XY** (arranged like the controller itself), **Grid**, or **List**.

Buttons are programmed and set up like everywhere else — see [Control > Button Settings](#).

Control › Footswitch Pedal



Everything on the floor: the M2's built-in footswitches, paired **F6W** / **F10W** pedals, and third-party pedals plugged into the TRS jacks.

From the bottom menu:

- **Pedal** — switch between footswitch pedals (e.g. **M2**, **TRS 1 : F3**, **F10W : 1**).
- **Layout** — depends on the pedal:
 - 10- and 6-button pedals: **Grid** or **List**.
 - **Universal** (resistor-ladder) pedals: pick any grid shape (5x2, 3x2, 2x2, ...) to match the pedal.
- **F2** / **F3** TRS pedals: **Swap L/R** instead, to flip the left and right switches.

Buttons are programmed and set up like everywhere else — see [Control › Button Settings](#).
Universal pedals are taught with **Learn** (see **Universal Footswitch Learn**).

Example: Custom layout with Universal Footswitch

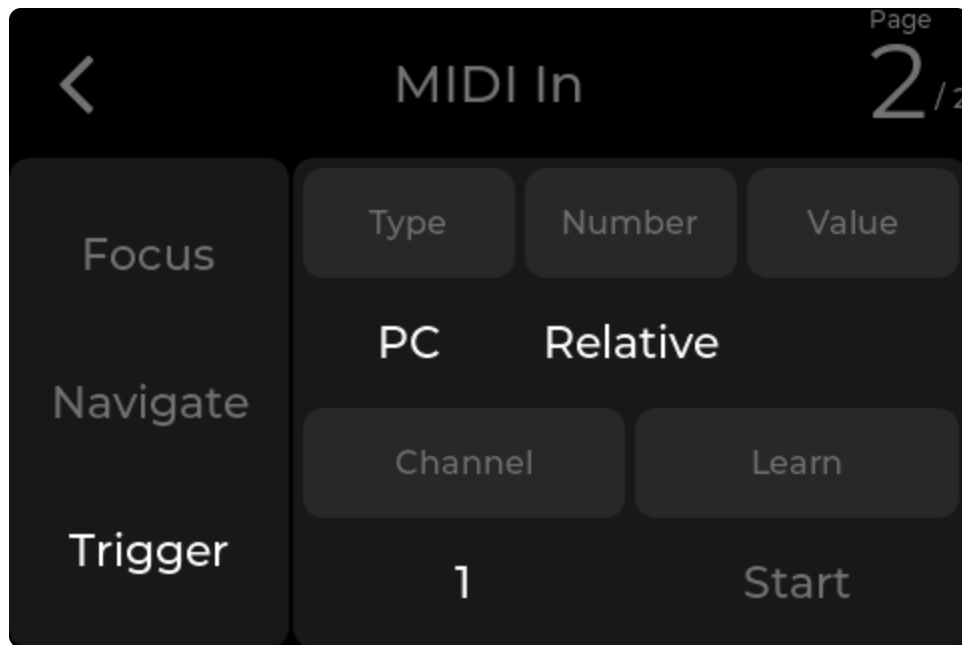


Control > MIDI In



Virtual buttons fired by **incoming MIDI Trigger** (5-pin, USB, or Bluetooth) instead of a physical press: **12 Pages × 10 buttons = 120**. Use it when another device is in charge — a DAW, a lighting desk, or another MIDI controller — and you want its PC / CC messages to run Graviton Actions.

Example: your keyboard sends PC **N** → Graviton recalls Preset **N**.



Each virtual button runs the same **Actions** as any other button — see [Control > Button Settings](#).

From the bottom menu:

- **Layout** — **Grid** or **List**.
- **Apply To** — whether the controller settings (**Global Bank**, **Focus**, ...) apply to **This Page** or **All Pages**.

Differences from physical buttons:

- Incoming MIDI has no "let go", so **Hold**, **Double Tap**, and **Release** aren't available.
- Receiving the same message again still steps through **2nd Tap** / **3rd Tap** (e.g. PC 1 again = **2nd Tap**).

Absolute Trigger

Fires on **one exact message** — a set Channel, Number, and Value (where they apply).

Relative Trigger

Set **Number** to **Relative**. Any message on the matching Channel / Value fires the Action, and the **number received** picks the target:

Action	Received number X means
Global Preset	Recall Preset X
Bank Preset	Recall the X-th Preset in the current Bank
Song Preset	Recall the X-th Preset in the current Song
Bank Jump	Go to Bank X
Song Jump	Go to Song X
Setlist Jump	Go to Setlist X

Use **Offset** to fix off-by-one numbering between devices — e.g. if receiving 10 recalls Preset 11, set **Offset** to **-1**.

Control > EC Controller

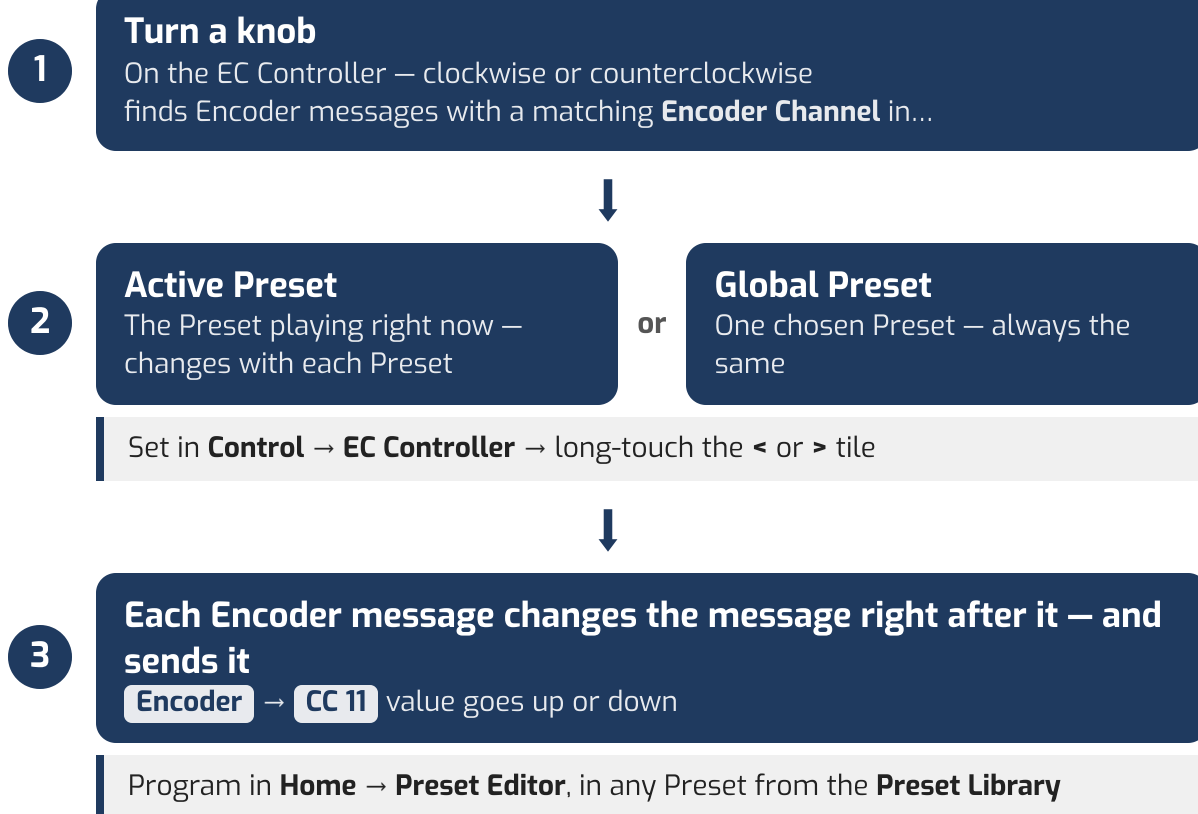


Turn a knob to change a value smoothly — delay mix, reverb level, gain, BPM — like an expression pedal you control by hand. Each **EC Controller** has **2 knobs**: turn them (clockwise / counterclockwise), or push them down to use them as **2 buttons**.

Example: left knob = delay mix, using a **Counter CC** so the mix doesn't jump when you change Presets. Push the knob to bypass the delay.

How it works

Knobs don't send MIDI directly. They drive **Encoder** messages that you place inside a Preset:



An **Encoder** message can control the message right after it:

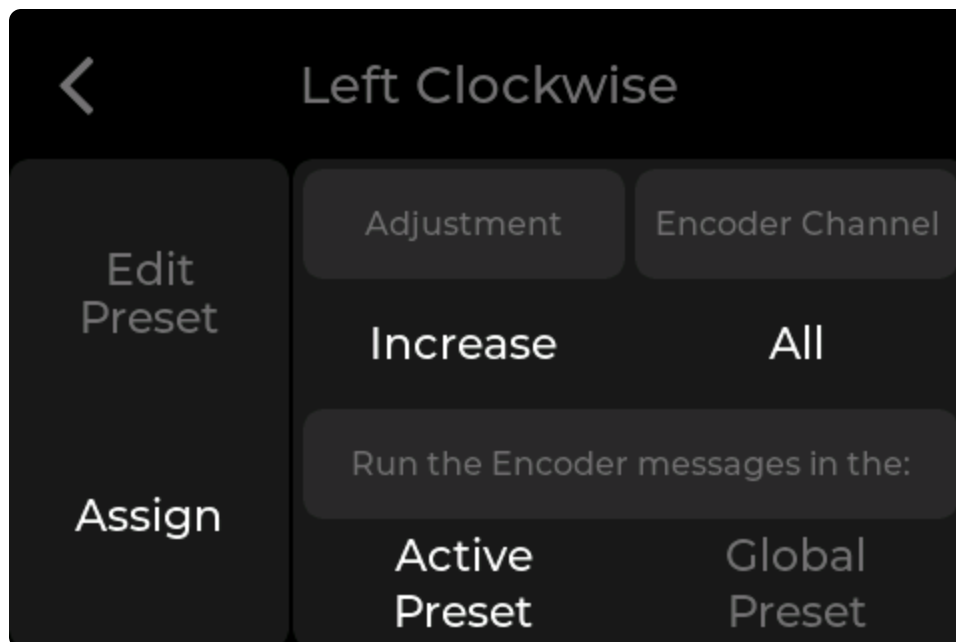
- **CC** and **PC**

- **Counter CC / Counter PC** (+ / - / Update / Send)
- **Control Voltage**
- **Clock** (BPM)
- **Pitch Bend**

Quick Start

1. Go to **Home** → **Preset Editor** and tap **+** to add a message.
2. Change its **Type** to **Encoder**. Leave **Encoder Channel** on **All** for now.
3. Tap **+** again and add a **CC** message right after it.
4. Turn a knob — the CC value changes on screen and is sent.

EC knobs settings



Each knob has three blocks: < (turn counterclockwise), + (push), and > (turn clockwise).

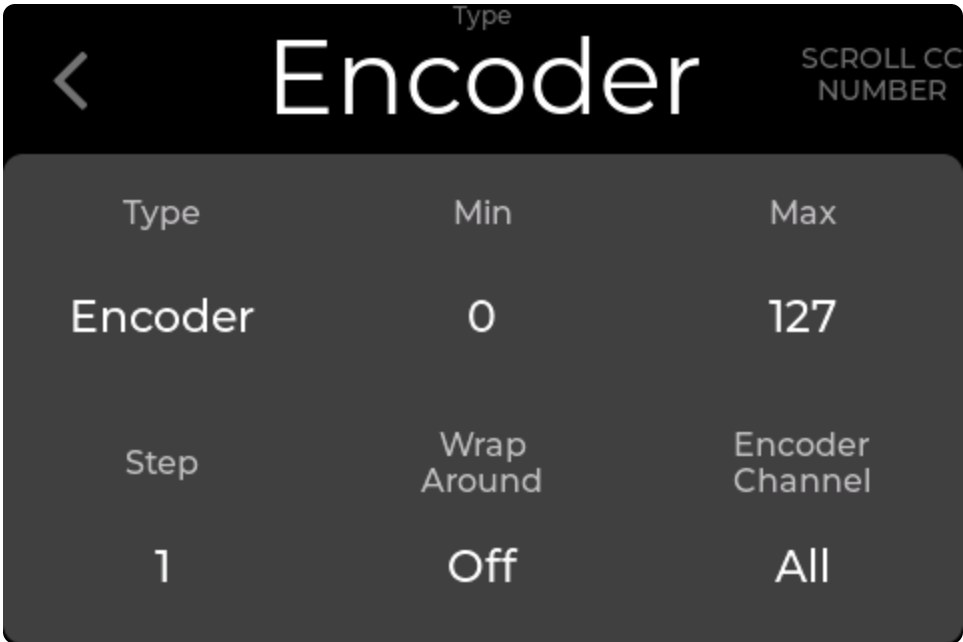
Long-touch < or > to set up that knob:

- **Adjustment** — **Increase** or **Decrease**.
- **Encoder Channel** — **All** (default) or **1–20**.
- **Run the Encoder messages in the:**
- **Active Preset** — the knob controls whatever the current Preset contains. E.g. reverb mix in Preset 1, gain in Preset 2.
- **Global Preset** — the knob always controls one chosen Preset, whatever Preset is active. Good for fixed jobs like master volume.
- **Sync** tab (< only) — by default < mirrors > in the opposite direction. You can assign it a different Action instead.

Long-touch + to assign an Action, like any other button — see [Control > Button Settings](#).

To switch between ECs, press any button on the EC you want, or use **Controller** in the bottom menu.

Encoder message settings



Setting	What it does
Min / Max	The range of values the knob can reach
Step	How much the value changes per click of the knob
Wrap Around	On: going past Max jumps back to Min (and vice versa). Off: stops at the end
Encoder Channel	Which knob(s) control this message — All , or 1–20

Encoder Channels work like MIDI channels. Each knob has an Encoder Channel, and each Encoder message has one too. Turning a knob only runs Encoder messages with the **same** channel. **All** on either side matches everything.

Advanced

Encoder + Counter CC / PC. Put an Encoder message in front of a **Counter CC** or **Counter PC** to get **Min**, **Max**, and **Wrap Around**, which a Counter doesn't have on its own. The Encoder's direction and **Step** replace the Counter's. The value is kept across Preset changes — ideal for global parameters.

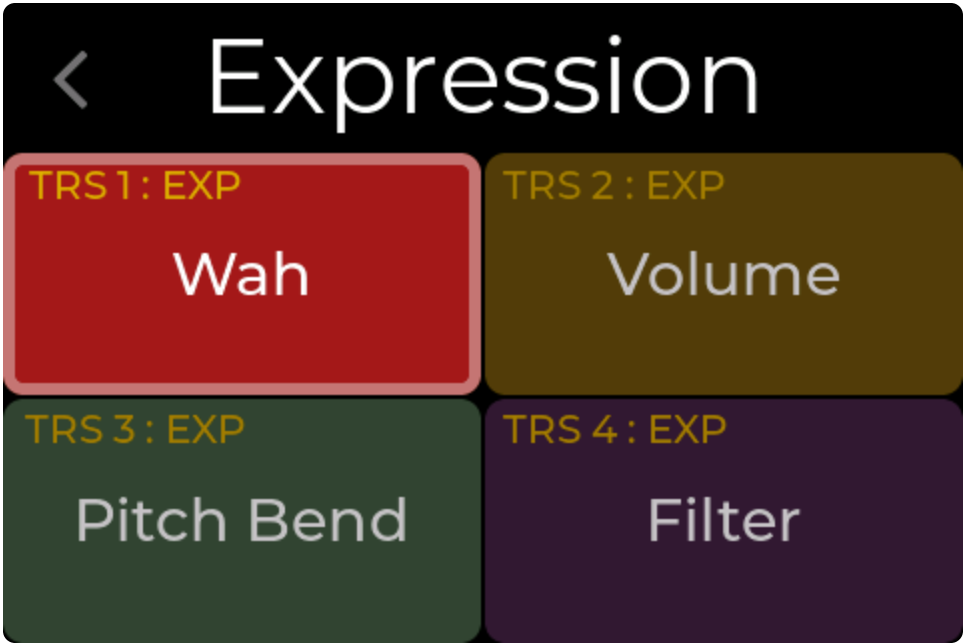
 *Note: The knob's Adjustment (Increase / Decrease) overrides the Counter's + / – direction.*

Stacking. Several Encoder messages in front of one message all control it. Give each a different **Min / Max / Step** — e.g. one knob for coarse and one for fine adjustment.

Scroll CC Number. When enabled on an Encoder message, the knob changes the **CC number** instead of the value, and **Min / Max / Step / Wrap Around** apply to the number. Stack it with a second Encoder to pick a parameter with one knob and adjust it with the other.

Buttons as knob clicks. Any button — a footswitch, an XY, or a **MIDI In** trigger — can act like one click of an EC knob. Assign it the **Encoder Click** Action (**Assign Action** → **Main Actions**, on **1st Tap** or **Hold** only). Then set the same options as an EC knob: **Adjustment**, **Encoder Channel**, and **Active Preset / Global Preset**. Use it to step a value up or down with your feet, e.g. one footswitch for **Increase** and another for **Decrease**.

Control > Expression Pedal



Sweep an Expression Pedal to change a value continuously — volume, wah, delay mix. There are two ways to set it up:

Method	Where you program it	What the pedal controls
Per Preset	Expression messages inside a normal Preset (Home → Preset Editor), mixed with other messages	Whatever the current Preset contains — changes with the Preset. Any pedal on any TRS jack controls it
Global	An Expression Preset (Control → Expression Pedal) — Expression messages only	The same Expression Preset, whatever Preset is active — assigned to one jack

⚠ *Note: If a jack has a Global Expression Preset and the current Preset has Expression messages, the pedal controls both. Stick to one method per jack, or switch deliberately with Assign Expression (below).*

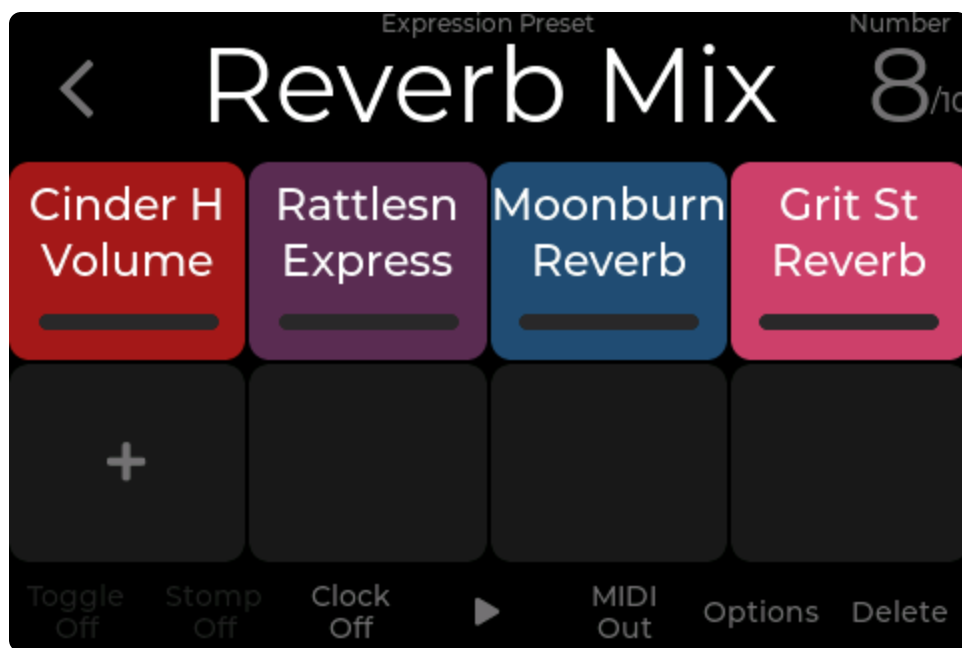
Assigning an Expression Preset

Control → **Expression Pedal** shows one tile per TRS jack set to **Expression**. There are two ways to assign an Expression Preset to a jack — use whichever you prefer:

- **From the Editor:** tap the jack's tile to open its current Expression Preset in the **Expression Preset Editor**, then tap the number at the top right to open the **Expression Preset Library** and pick a different one.
- **Directly:** long-touch the jack's tile → **Change Preset** to go straight to the **Expression Preset Library** and pick one. (**Edit Preset** in the same menu does the same as tapping the tile.)

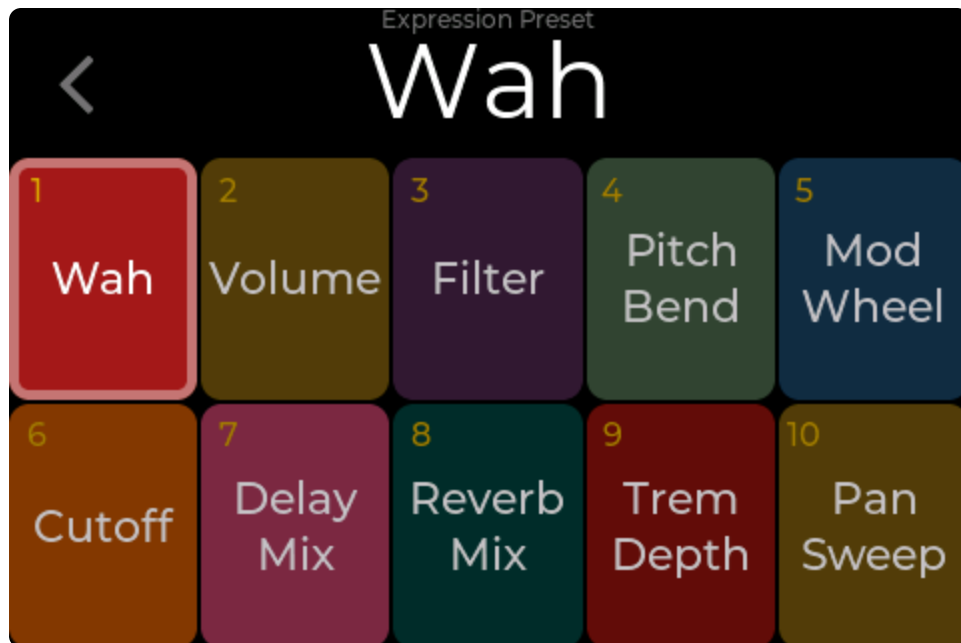
To change it automatically, add an **Assign Expression** message to a normal Preset. Pick the **TRS jack** (1–2 on M1, 1–4 on M2) and the Expression Preset. When that Preset is sent, the jack switches to the new Expression Preset.

Expression Preset Editor



Works like the **Preset Editor**, but only holds Expression messages. Values update on screen live as you move the pedal. Tap **x / 10** at the top right to open the **Expression Preset Library**.

Expression Preset Library



10 Expression Presets, shared by all pedals in both **Bank Mode** and **Song Mode**.

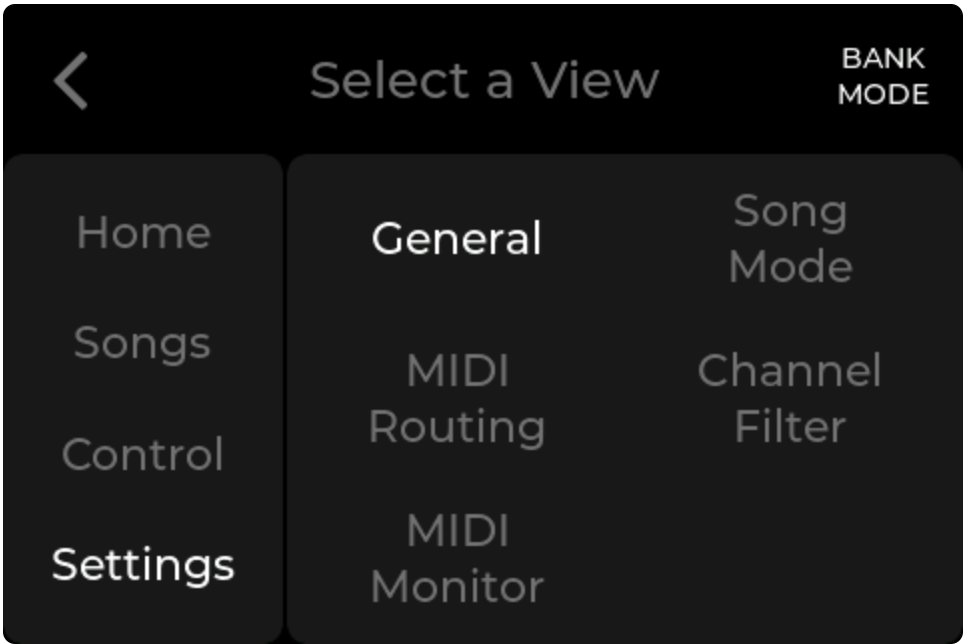
- **Expression Preset Name** — tap to rename.
- **Expression Presets 1–10:**
- **Single touch** — assign it to the current pedal.
- **Double touch** — open it in the **Expression Preset Editor**.
- **Long touch** — **Edit / Rename / Copy / Swap / Reset Preset** (clears it).

 *Note: The Editor and Library always work on the pedal you opened them from. Moving a different pedal makes it the current one, and the screen follows. To pick a pedal on purpose, go back to Control → Expression Pedal and tap its tile.*

Good to know:

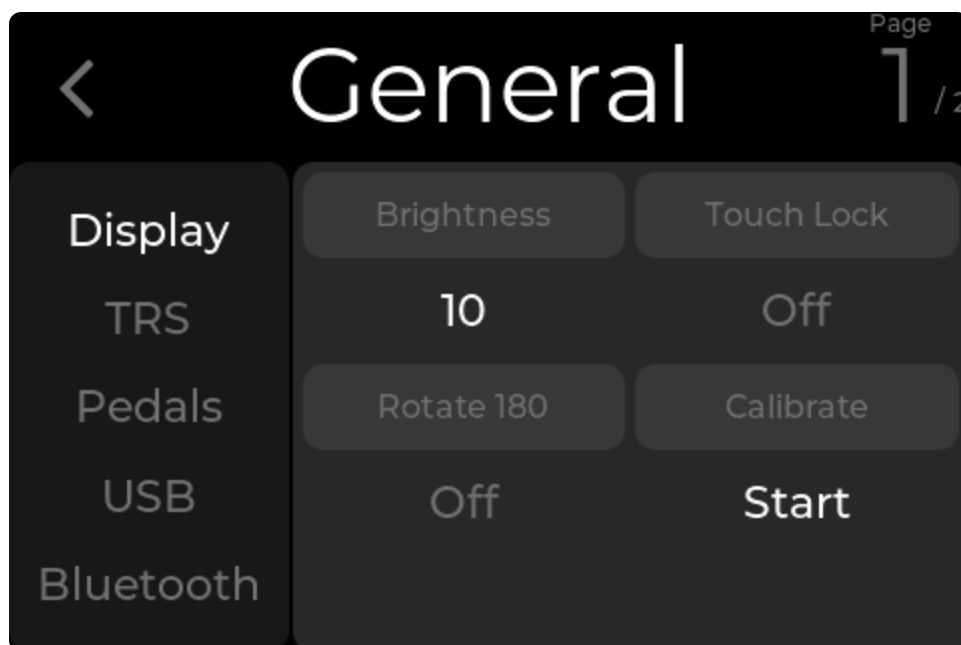
- You can assign an Expression Preset to an empty jack and plug the pedal in later.
- Changing a TRS jack's function doesn't delete its Expression Preset.
- When you change Presets, the pedal's current position is sent, so the new sound starts where your foot already is.

Settings



Screen	What it covers
General	Display, TRS, Pedals, USB, Bluetooth (page 1); Pairing, Timing, Clock, Data, Update (page 2)
Song Mode	What happens at the end of a Song / Setlist; Preset reset on Song change
MIDI Routing	Out / In / Through / Clock for each interface
Channel Filter	Which MIDI channels each port may send (Out) or forward (Thru)
MIDI Monitor	Live log of incoming MIDI, for troubleshooting

General



Display

- **Brightness:** LCD brightness, from **1** to **10**.
- **Touch Lock:** After **5** / **10** / **30** seconds of no touch (or **Off**), the screen locks so a stray touch cannot change anything. Footswitches, XY/EC, and other controllers keep working. When locked, the back button turns **red** — **double-tap** it to unlock.
- **Rotate 180:** Flip the display **180** degrees (upside down).
- **Calibrate:** If touch accuracy drifts, tap **Start**, then tap the red squares until they turn white.

TRS

Choose what each TRS jack does.

- **TRS Port:** Tap to choose the TRS jack.

Mode:

- **MIDI Out:** Send MIDI to other gear. Pick the **Type** that matches the device's TRS wiring (if you're unsure, try each type).
- **Footswitch:** A wired footswitch pedal.
- **Expression:** A standard expression pedal.
- **Relay Out (via RV):** Connect to an **RV Adapter's Relay In** (switching through the adapter).
- **CV Out (via RV):** Connect to an **RV Adapter's CV In** (control voltage through the adapter).
- **Switch Out:** Connect directly to another pedal's AUX, tap-tempo, or channel-switch jack.

MIDI Out Type:

- **Type A:** Tip: Signal, Ring: Power (the MIDI standard).
- **Type B:** Ring: Signal, Tip: Power.
- **Tip Active:** Tip: Signal, Ring: Floating.
- **Ring Active:** Ring: Signal, Tip: Floating.

Footswitch Type:

- **F10/W:** Luminite **F10** or **F10W** in wired mode.
- **F6/W:** Luminite **F6** or **F6W** in wired mode.
- **1-button / 2-button / 3-button:** Common TRS footswitches; the layout is chosen automatically.
- **Universal:** A generic resistor-ladder pedal — use **Learn** (see Universal Footswitch Learning).

Calibration, debounce, and expression curves are in **General > Pedals**, once **TRS** has set the **Mode / Type**.

⚠ **Switch Out** (switching directly over a TRS jack) is meant for low-voltage pedals (5V or less), which includes most digital effects from brands like Strymon, Boss, Chase Bliss, and Eventide. It may pick up **noise from ground loops**.

Analog pedals and amp channel-switching jacks often carry more than 5V, which can damage the TRS jacks. Use the M2 Relay port or an RV Adapter instead — they switch **any** voltage and eliminate ground loops.

Pedals

Fine-tune each footswitch and expression pedal — the M2's built-in footswitches and any TRS jack set to **Footswitch** or **Expression** in **TRS**.

- **Pedal:** Tap to pick the pedal.

Footswitches (M2 built-in, **F10/W**, **F6/W**):

- **Calibrate → Start:** Fixes misaligned footswitches (e.g. pressing button 2 triggers action 3). Follow the on-screen steps to re-learn each switch.
- **Debounce:** Raise it (roughly **1–8**) if presses feel glitchy or double-fire. Higher filters more, but registers presses slightly slower.

Digital footswitches (**1/2/3-button**) have nothing to calibrate, so they only show **Debounce**.

Expression pedals:

- **Calibrate → Start:** Detects the plugged-in pedal type (potentiometer value, hot tip, or hot ring) and sets it up. Sweep the pedal fully when prompted.
- **Options:** Opens a popup with:

- **Noise Filter:** Reduces value jitter while the pedal is still. Higher values reduce jitter more but can feel less smooth.
- **Toe / Heel: Normal** or **Reversed** — flips which end of the sweep sends the higher value.
- **Response Curve: Linear** (default), **Log** (simulate a volume pedal's curve on a linear pedal), or **Anti-Log** (flatten a volume pedal's curve so it behaves more linearly).

USB

The Graviton M1/M2 can act as either a USB Device or a USB Host. (A USB Device can only connect to a USB Host, and vice versa.)

- **Device** (USB Type-C port): Connect the M1/M2 to a computer, phone, or tablet (these are always USB Hosts), or any other device with a USB Host port.

In this mode the M1/M2 can be powered and connected with a single USB cable (bus-powered). Make sure the host port can supply enough power (1A or more).

- **Keyboard On/Off:** Turns the Graviton M1/M2 into a virtual USB keyboard. Use it with **Keystroke** messages to control DAWs, computers, and iOS devices — especially DAWs with lots of keyboard shortcuts, such as Ableton Live.
- **Custom Name:** Change the USB Device name shown on the host computer. Useful when several Graviton M1/M2 units connect through a USB hub.
- **Host** (USB Type-A port): Connect other USB MIDI devices (e.g. pedals, MIDI controllers).

Host mode needs the M1/M2 powered from 9V. It can also power the connected device through its USB-A port, so make sure the 9V adapter can supply enough current for both.

 *Note: USB MIDI doesn't work with both the USB (Type-C) and USB Host (Type-A) ports plugged in. Connect only one at a time and set the USB mode to match.*

Bluetooth

The Graviton M1/M2 can take either the Bluetooth **Peripheral** or **Central** role. (A Peripheral can only connect to a Central, and vice versa.) A Peripheral broadcasts its presence (advertising); a Central scans for advertising Peripherals and starts the connection (pairing).

The M1/M2 reconnects automatically after a power cycle in both roles, as long as the other device supports it too.

- **Mode: Off / Peripheral / Central. Off** disables BLE (**Restart to take effect ?**).
- **Peripheral:** Connect to computers, phones, tablets, or Bluetooth MIDI devices in the Central role.
- **Advertising:** Whether the Graviton M1/M2 is visible to unpaired Bluetooth devices when they scan.
 - **Off:** Not visible to other Bluetooth scanners.

- **Auto** (default): For connecting to a **single** Central device. Advertising stops after a successful connection and resumes if disconnected.
- **Always On**: For connecting to **multiple** Central devices. Switch back to **Off** afterward to avoid unwanted connections from unpaired devices.
- **Custom Name**: Change the advertising name.

In Peripheral mode the M1/M2 can pair with up to 3 Central devices. Pair from each Central device separately.

- **Central**: Connect to other Bluetooth MIDI devices that are advertising (in the Peripheral role).
- **Scanning**: Scan for nearby advertising Bluetooth MIDI devices. If a device asks for a PIN code, enter it before connecting.
- **Paired**: Lists paired devices (up to 3). You can connect or disconnect them temporarily, or unpair them.

The M1/M2 remembers the devices in this list and reconnects to them after a power cycle.

Connection guide:

- **CME WIDI**:
- Option 1: Set the M1/M2 to **Peripheral** mode and unpair the CME device from anything it was paired with before. They connect automatically.
- Option 2: Set the M1/M2 to **Central** mode and unpair the CME device from anything it was paired with before. Then **Scanning** → **Start** → tap the device name in the list to connect.

⚠ **Note:*** For 1-to-1 MIDI connections, **you can use** either option. For CME WIDI Groups, **use** Option 1. For other brands or custom MIDI groups, **try** either option* *and see which works best for your setup.*

- **macOS**:
- Set the M1/M2 to **Peripheral** mode.
- On your Mac, open **Audio MIDI Setup** → **Window** → **Show MIDI Studio**, click the button with the **Bluetooth** logo → select **Graviton M1/M2 BLE** → **Connect**.
- Then select **Graviton M1/M2 BLE** as the MIDI input/output in your DAW or app.

⚠ **Note:** The M1/M2 won't appear in the system Bluetooth settings like regular devices such as headphones or speakers.

- **iOS / Android**:
- Set the M1/M2 to **Peripheral** mode.
- On the phone or tablet, scan and connect from the Bluetooth or settings page of the **app you're using**.

⚠ **Note:** The M1/M2 won't appear in the iOS or Android system Bluetooth settings like regular devices such as headphones or speakers.

- **Windows:**
- Set the M1/M2 to **Peripheral** mode.
- Windows 10: **Devices** → change **Devices discovery** to **Advanced** → **Add Device** → **Graviton M1/M2 BLE**.
- Windows 11: **Settings** → **Bluetooth & devices** → **Add Device** → **Show more** → **Graviton M1/M2 BLE**.
- Once paired, you need a UWP app from the Microsoft Store to connect and route MIDI to your software, for example:
 - **MIDIberry BLE** — connects to BLE MIDI devices.
 - **BLE MIDI Tester** — for testing the connection.

⚠ **Note*:** BLE MIDI devices like Graviton M1/M2 will appear as "Not connected" or "Disconnected" in Windows Bluetooth Settings if no UWP (Universal Windows Platform) app is actively using them. This is normal behavior — the BLE MIDI connection is app-driven*, not system-level like classic Bluetooth audio or HID.

- **Other Bluetooth MIDI devices** (advertising):
- Set the M1/M2 to **Central** mode.
- **General > Bluetooth > Scanning** → **Start** → tap the device name in the list to connect.

Pairing

Pair Luminite wireless controllers here (Bluetooth MIDI pairing is under **Bluetooth**). Tap **XY / EC** or **F10W / F6W**, tap an empty slot, choose the type if asked, and follow the on-screen steps.

- **XY / EC:** Up to 10 XY/EC Controllers — handy if you play several instruments, each with its own controller.
- **F10W / F6W:** Up to 10 wireless footswitches. Tap an empty slot → **Pair a new Wireless Footswitch** → choose **F6W Footswitch** or **F10W Footswitch** → hold the leftmost top and bottom switches as prompted → **Paired Successfully**.
- **Channel > Change:** If interference causes wireless dropouts, try another channel (**Change Pairing Channel**). All paired XY/EC Controllers and wireless footswitches must be re-paired after a channel change.

Timing

Button timing, in milliseconds:

- **Double Tap:** The longest gap between two presses that still counts as a double tap.
- **Hold:** How long a button must be held to trigger its **Hold** action.

- **Hold Repeat:** While a button is held, how often its action repeats (**300–2000 ms**, default **1000 ms**; **Off** = no repeat).
- **Send Delay:** The short gap between messages whenever Graviton sends a sequence (e.g. a Preset, or an expression sweep) — in effect, a global sending rate.

Increase **Send Delay** if the receiving device misses messages because they arrive too fast.

⚠ *Note: If the value is high and a Preset with many messages is sent, the M1/M2 may seem to freeze briefly — it finishes sending before doing anything else.*

Footswitch debounce and calibration are in **General > Pedals**.

Clock

While Switching Presets:

- **Keep All Clocks Running** (default): Clocks keep running, unless the new Preset contains a Clock stop message.
- **Stop All Clocks:** Clocks stop as soon as you switch to another Preset.

Analog Tap Clock Cycle:

- **4 Taps:** Send 4 taps, then stop (for pedals that only need 4 taps to set their BPM).
- **Continuously:** Keep sending Analog Clock, like MIDI Clock.

Data

Save all your data — **Presets, Songs, Setlists, Settings**, and more — into a **Backup**. You can keep up to 10 **Backups** (e.g. one per rig) and switch between them.

- **Backup:** Opens the Backup screen with 10 slots:
- **Save / Load:** Save everything into a slot, or load a slot back — switch between rigs in seconds.
- **Import / Export:** Copy a slot to or from another device over Wi-Fi — move a rig to a second unit, or keep a copy on your computer.
- **Reset Data:**

Choice	Confirm text (summary)	What's kept
Settings	Reset Settings to Default — everything else preserved	Presets, Songs, Setlists, Actions, Backups, and footswitch / expression calibration
All	Reset All — Backups will be preserved	Backups only — everything else is erased

Each reset asks you to tap **Confirm** more than once.

- **Hardware factory reset** (pin hole on the unit): Restores the factory firmware and **erases Backups** too. Use it only as a last resort — **Reset All** is enough in most cases. See [Graviton M1](#) / [Graviton M2](#).

Update

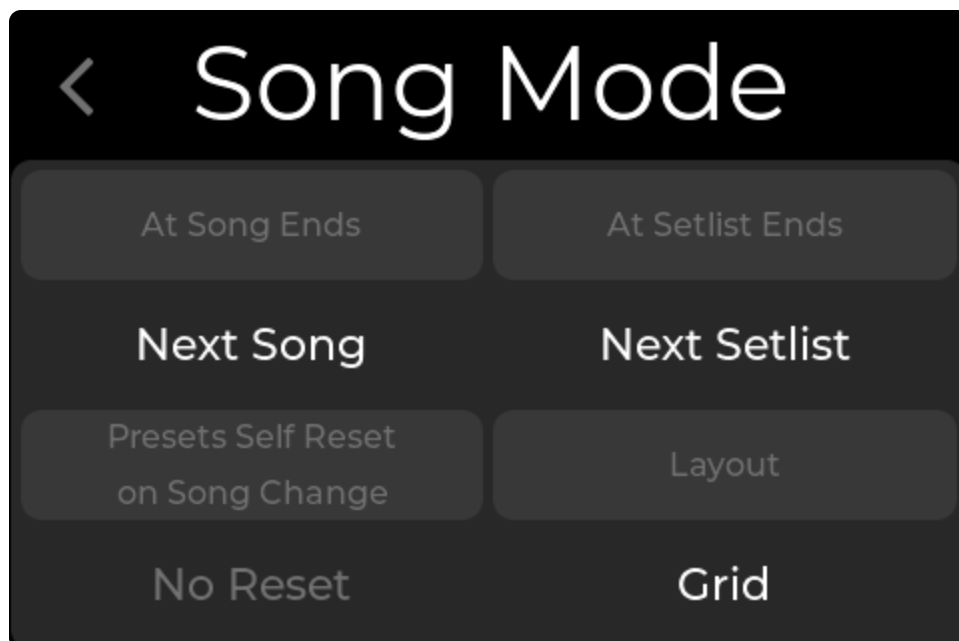
- **Update / Rollback:** Check for and install an over-the-air software update over Wi-Fi (needs internet access). No computer needed.
- **Latest Beta:** Try the newest features before their official release.
- **Latest Stable:** Tested and recommended for regular use.
- **Previous Stable:** Go back to this if the latest **Stable** introduced a bug.
- You can switch between **Beta** and **Stable** at any time without losing your data.
- For gigging, stay on **Stable**.

If an update fails — possible causes and fixes:

- **Weak Wi-Fi signal:** Move the M1/M2 closer to the router and try again.
- **No access to the router:** Connect the M1/M2 to your phone's hotspot instead.
- **Weak or noisy power supply:** A poor 9V/USB adapter can affect Wi-Fi reception. Try a different power source.
- **Interference nearby:** Try updating somewhere else if possible.

⚠ **Note:** If you're updating from pre-4.6.5 and the update gets stuck in a crash loop, try turning off any connected Bluetooth devices. This was fixed in 4.6.5 and later.

Song Mode

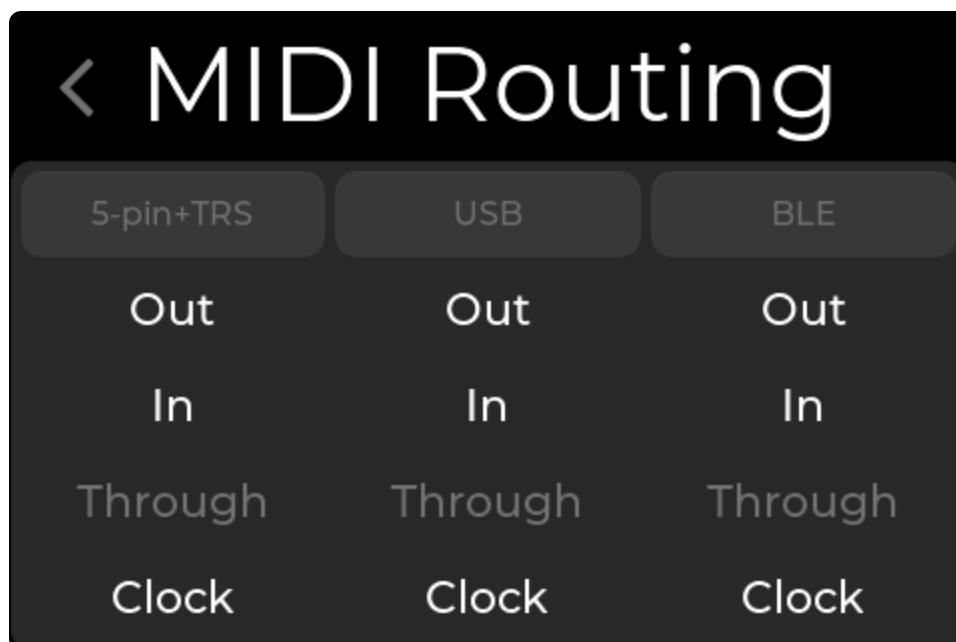


These settings only apply in **Song Mode**, not **Bank Mode**.

- **At Song Ends** — what happens when you scroll past the end (or beginning) of the current Song:
- **Off**: Nothing.
- **Wrap Back**: Jump to the first (or last) Song Preset.
- **Next Song**: Jump to the next (or previous) Song.
- **At Setlist Ends** — what happens when you scroll past the end (or beginning) of the current Setlist:
- **Off**: Nothing.
- **Wrap Back**: Jump to the first (or last) Song.
- **Next Setlist**: Jump to the next (or previous) Setlist.
- **Presets Self Reset on Song Change** — what happens to Preset **Toggle** / **Rotation** states when you load another Song:
- **No Reset**: States stay as they are.
- **Self Reset to On**: All Presets in the Song reset to **On**.
- **Self Reset to Off**: All Presets in the Song reset to **Off**.
- **Self Reset to On+Send**: Reset to **On** and send the On messages.
- **Self Reset to Off+Send**: Reset to **Off** and send the Off messages.
- **Layout**: **Grid** or **List** for **Playing Song**.

Each Song's own **Song BPM** and **Startup Preset** are set on [Songs > Playing Song](#) → **Song Options**.

MIDI Routing



The grid has **three columns**. Each column is a **port family** — one switch for a group of related jacks, not a single plug:

Column (UI)	Port family — what it covers
5-pin+TRS	Wired MIDI: 5-pin DIN MIDI In / Out , plus every TRS jack set to MIDI Out or MIDI Out Only (M1 : TRS 1–2, TRS 3 1/8 Out; M2 : TRS 1–4, TRS 5 1/8 Out). On the M2 , the 1/8" MIDI In jack is in this family too (use either it or 5-pin DIN In, not both).
USB	USB Device (Type-C) or USB Host (Type-A) — whichever is active (only one at a time).
BLE	Bluetooth LE MIDI (Peripheral or Central role).

Each column has four buttons — **Out**, **In**, **Through**, and **Clock**. Tap one to **enable** it; tap it again to disable it. Each button applies to **the whole family**. For example, disable **5-pin+TRS > Out** and Graviton stops sending on 5-pin **and** on every TRS MIDI Out jack.

- **Out** (enabled by default): Graviton **sends its own** messages on that interface (Presets, Actions, Expression, Tracking, etc.). Disable it to mute that path without changing any Presets.
- **In** (enabled by default): Graviton **receives** MIDI on that interface (MIDI In Actions, Tracking feedback, SysEx capture, etc.). Incoming MIDI must arrive on a column with **In** enabled before anything else can act on it — including **Through**.
- **Through** (disabled by default): MIDI thru — forwards *incoming* MIDI (from any interface with **In** enabled) out of every interface that has **Through** enabled.
- **Clock** (enabled by default): MIDI Clock can be sent on that interface. The BPM comes from the Preset Editor's BPM keypad or a Preset's **MIDI Clock** message; **General > Clock** decides whether clocks keep running when you switch Presets. An external clock is forwarded by **Through**, not by this button — real-time ticks pass through whenever **Through** is enabled.

Through

- **Source and destination can differ.** Example: a PC arrives on **USB (In enabled)**. If **5-pin+TRS > Through** is enabled, it's forwarded out the 5-pin / TRS MIDI Out jacks. If **BLE > Through** is also enabled, it's forwarded over BLE too.
- **Graviton's own messages don't use Through** — they always go out through **Out**. **Through** only forwards what arrives from outside.
- **What's forwarded:** channel messages (Note, CC, PC, Pitch Bend, Aftertouch, ...), SysEx, Song Select / Song Position, Tune Request, and real-time messages (MIDI Clock, Start / Stop / Continue, Active Sensing, System Reset).
- **Typical use:** Graviton sits mid-chain. Example: a laptop on **USB** sends clock and PCs → Graviton → delay and amp on 5-pin. Enable **USB > In** and **5-pin+TRS > Through** so the pedals still hear the laptop. Graviton's own Presets still go out through **Out**.
- **Watch for loops:** enabling **Through** on the same interface that feeds **In** (especially USB Host or BLE) can echo messages back to the source. Only enable **Through** on the paths that should carry MIDI onward, away from Graviton.
- To limit which MIDI channels are sent by **Out** / **Through** on each port, use [Channel Filter](#) (below).

Channel Filter



Choose which **MIDI channels (1–16)** each port may send (**Out**) and forward (**Thru** — the same as **Through** on [MIDI Routing](#)).

Example: your amp on TRS 1 listens on channel 1 and your delay on TRS 2 on channel 2. Allow only channel 1 on TRS 1 and only channel 2 on TRS 2, so each Program Change reaches only the device it's meant for.

Each column is a port or a group of jacks that share one filter, listed left to right:

- **M1: TRS 1, TRS 2, TRS 3 / 5-pin, USB, BLE.**
- **M2: TRS 1+2 / 5-pin / 3.5mm, TRS 3, TRS 4, USB, BLE.** TRS 1, TRS 2, the 5-pin jack, and the 3.5mm jack share one filter.

Each column has an **Out** and a **Thru** button. Tap one to open its channel list, then tap a channel number (**1–16**) to allow or block it — allowed channels **highlight**. Tap **ALL** (top-right) to allow every channel at once. Back on the grid, an **Out / Thru** button highlights when at least one channel is allowed.

In isn't set here — incoming MIDI is controlled from **MIDI Routing**.

MIDI Monitor



#	Port	Type	Number	Value	Channel
1	5-pin	CC	7	100	1
2	USB	PC	5		1
3	BLE	Note	60	127	1

MIDI Monitor is the "is anything even arriving?" screen, useful for troubleshooting. It shows a live view of incoming MIDI from active interfaces (**5-pin**, **USB**, **BLE**).

Supported Message Types

- Note On / Off
- Control Change (CC)
- Program Change (PC)
- Pitch Bend
- SysEx (System Exclusive)

⚠ **Real-time** messages (MIDI Clock, Active Sensing, ...) are hidden here to keep the list readable. Clock still runs on the ports — you just don't see every tick.

Columns:

- **#**: Message order
- **Port**: Source (**5-pin**, **USB**, or **BLE**)
- **Type**: Note On, CC, SysEx, ...
- **Number**: Note or CC number, etc.
- **Value**: Velocity, CC value, ...
- **Channel**: 1–16

SysEx shows a short hex preview in **Value** — tap it for the full message.

Safe Mode / Boot Issue

If the unit keeps crashing and restarting on its own (a crash loop), the next startup shows a **Boot Issue** screen instead of the normal Home view.

- **Continue:** Starts the unit in **Safe Mode** — Bluetooth off, USB Host off — so you can still use it and get your data out. Export a backup **before** turning Bluetooth or USB Host back on, in case they're what's causing the crash.
- **More Options:**
- **Import Export:** Send a backup off the unit over Wi-Fi (same as **General > Data > Import / Export**).
- **Reset Settings:** Restores default settings only. Presets, Songs, Setlists, Actions, and Backups are kept (same as **General > Data > Reset Data > Settings**).
- **Reset All:** Full user-data wipe. **Backup** slots are kept (same as **General > Data > Reset Data > All**).

Use **Continue** first and export a backup while you still can. Only use **Reset Settings** or **Reset All** if a normal boot keeps failing and Safe Mode doesn't fix it.

Chapter D: Glossary

Common Terms

- **Controllers:** Devices that control the Graviton M1/M2 — the **XY/EC Controller**, a **Footswitch Pedal** (the M2's built-in footswitches, an **F10/W**, an **F6/W**, etc.), or the **MIDI In** port, which triggers Actions from incoming MIDI CC/PC messages.
- **Buttons:** Physical switches on the controllers above, or MIDI messages received through **MIDI In**. Each button can be assigned multiple Actions — **1st Tap**, **2nd Tap**, **3rd Tap**, **Hold**, **Double Tap**, and **Release**.
- **Preset messages:** The messages programmed inside a Preset (see **Home** → **Preset Editor**). Can be MIDI (**CC**, **PC**, **Note**, **SysEx**, etc.) or non-MIDI (**CV**, **Relay**, etc.).
- **Button actions:** Messages assigned to a button under **Control** (**XY / EC / Footswitch / Expression / MIDI In**). Can be **Bank Preset/Global Preset**, **Bank +/-**, **Song +/-**, etc.
- **Presets:** A collection of up to 16 Preset messages, organized into 12 Banks of 10 Presets each — see **Home** → **Preset Library**. Presets are shared between all controllers.
- **Current Preset:** The Preset most recently sent (recalled).
- **Toggle:** Latching On/Off (or Rotation) behavior on a Preset, so one button sends a different message each time depending on its current state.
- **Bank:** A group of 10 Presets.
- **Song:** A collection of up to 10 Presets, picked from the Preset Library.
- **Song Preset:** A Preset that's part of a Song.
- **Setlist:** A collection of up to 10 Songs, picked from the Song Library.
- **Playing Mode:** **Bank Mode** or **Song Mode**. Switch it from **Select a View** (top-right **BANK MODE / SONG MODE** → popup **Playing Mode**). Button Actions are stored separately for each mode.
- **Expression Message:** A message type whose value tracks an expression pedal's position continuously, rather than a fixed value.
- **Expression Preset:** A group of up to 6 Expression Messages.
- **TRS (Tip, Ring, Sleeve):** A stereo jack, available in either ¼" or ⅛" sizes.

Preset messages

Tap **Type** in **Preset Editor** → **Select a Type**. Use **Page 1 / 2** in the title bar.

Page	Tab	Choices on that tab
1	MIDI	CC , PC , CC (category), PC (category), Note (category), SysEx
1	More	Clock (category), Song (category), Preset Chain (category), Special (category), Switch , Control Voltage
1	EXP EC	Expression (category), Toe/Heel (category), Assign Expression , Encoder
2	Bank	Bank + / - , Preset + / - , Bank Jump
2	Song	Song Preset (category), Setlist Library (category), Song Library (category), Song Jump
2	Tap Marker View	Tap Marker , View Jump

Page 1 — MIDI, More, EXP EC

- **MIDI** (tab) →
- **CC**: Send a **Control Change**.
- **PC**: Send a **Program Change**.
- **Counter CC and PC with + / - / Jump**: A normal **CC** / **PC** always sends the exact value you stored. **Counter CC** and **Counter PC** work differently, in two ways:
 1. **They remember their last value**. Each **+**, **-**, or **Jump** message moves a running counter instead of sending a fixed number, so the value keeps climbing (or jumps) every time you trigger it—it doesn't reset back to where it started.
 2. **They sync across Presets**. Think of every **Counter CC** (or **PC**) message with the same Channel/Number as **copies of one shared message**, mirrored across however many **Presets** use it—nudge the value from any one of them, and every copy updates together. **Counter CC** applies this to a CC value. **Example**: give the delay-mix **Counter CC** to several **Presets** (verse, chorus, solo)—the current value is retained and synced between them, so bumping the mix up on one carries over no matter which of those **Presets** you switch to next. Pair it with an **EC** encoder or **Expression** to sweep a global level like reverb, delay, or gain. **Counter PC** applies the same idea to program numbers. **Example**: use the same **Counter PC** as a patch scroller in different **Presets**—the current patch number is always displayed correctly. Each **Counter CC** / **Counter PC** message does one of three things: - **+**: Step the counter up by **Step**, then send. - **-**: Step the counter down by **Step**, then send. - **Jump**: Tap **Mode** to choose: - **Jump+Send**: Set the counter to the **Jump** target, then send. - **Jump**: Set the counter to the **Jump** target, don't send. - **Send (Current Value / Current Number)**: Send the current counter, don't change it.
- **Note** (category, **MIDI Note Type**) →
 - **Note On / Note Off**: MIDI note with velocity.
 - **All Note Off**: All notes off on the channel.
 - **Pitch Bend**: Pitch-bend message.
 - **Poly Aftertouch / Channel Aftertouch**: Aftertouch messages.

- **SysEx: SysEx Library** entry (up to **64** bytes; split longer data as ordered messages **F0 ... F7**).
- **More** (tab) →
- **Clock** (category, **MIDI Clock Type**) →
 - **MIDI Clock**: Start or set Clock at a BPM (**MIDI Routing > Clock**); overrides **global BPM** when this Preset runs.
 - **BPM +**: Nudge **global BPM** up by the configured step.
 - **BPM -**: Nudge **global BPM** down.
 - **Start Stop Continue**: MIDI transport Start / Stop / Continue (not the continuous MIDI Clock tick stream).
 - **Tap Tempo**: Set **global BPM** from taps; **Clock Duration Continuous** or **1–11 s**.
 - **Analog Clock**: Tap-tempo pulses at a BPM (**General > TRS** → **Switch Out; Analog Tap Clock Cycle** in **General > Clock**).
- **Song** (category, **MIDI Song Type**) →
 - **Song Select**: Standard MIDI Song Select (not Graviton **Songs**).
 - **Song Position Pointer**: Standard MIDI SPP.
- **Preset Chain** (category) →
 - **Preset Chain**: Run another library Preset when this message plays. **Mode**:

Mode	Behavior
Send Preset	Trigger the chained Preset (also flips Toggle if enabled)
Send On State / Send Off State	Send that state's messages
Send Current State	Send the target's current On/Off state
Toggle On/Off	Flip On/Off without sending
Turn On / Turn Off	Set state without sending

- **Set Toggle: On, Off, or Flip** another Preset's toggle state (no messages sent).
- **Special** (category) →
- **Keystroke**: Send a USB keyboard key press (**Settings > USB > Keyboard On**).
- **Pause**: Wait **0–382 ms** before the next line (default **30**).
- **A/B**: Messages before **A/B = On**; after = **Off (Toggle Mode / Momentary)**.
- **Switch**: Set relay / TRS switch state (**M2** or **RV Adapter**).
- **Control Voltage**: Output a CV level (**M2 CV** or **RV Adapter**).
- **EXP EC** (tab) →
- **Expression** (category, **Expression Type**) →
- **Expression CC**: CC follows pedal position while moving.
- **Expression Clock**: **MIDI Clock** BPM follows pedal position.
- **Expression CV**: CV output follows pedal position.
- **Expression Pitch Bend**: Pitch bend follows pedal position.
- **Toe/Heel** (category, **Toe / Heel Type**) →

- **Toe Down CC / Heel Down CC**: One-shot CC at full toe or heel.
- **Toe Down PC / Heel Down PC**: One-shot PC at full toe or heel.
- **Toe Down Preset / Heel Down Preset**: Fire a **Preset** at full toe or heel.
- **Assign Expression**: Attach an **Expression Preset** to a **TRS** expression jack.
- **Encoder**: Used internally for **EC** encoder input (see **Encoder**).

Page 2 — Bank, Song, Tap Marker View

- **Bank** (tab) →
- **Bank +**: Go to the next bank.
- **Bank -**: Go to the previous bank.
- **Preset +**: Go to the next Preset in the bank.
- **Preset -**: Go to the previous Preset in the bank.
- **Bank Jump**: Jump to a chosen bank (1–12).
- **Song** (tab) →
- **Song Preset** (category) →
 - ← → ↑ ↓: Move the active Song Preset slot on **Playing Song**.
 - **Position**: Pick the active slot (**Position** picker).
 - **1st–10th, Default (Select a Song Preset)**: Jump to that slot in the current **Song**.
- **Setlist Library** (category) →
 - **Song(S) +**: Next song in the **Playing Setlist**.
 - **Song(S) -**: Previous song in the **Playing Setlist**.
 - **Setlist +**: Next **Setlist** in **Setlist Library**.
 - **Setlist -**: Previous **Setlist**.
 - **Setlist Jump**: Jump to a chosen **Setlist**.
- **Song Library** (category) →
 - **Song(L) +**: Next song in **Song Library**.
 - **Song(L) -**: Previous song in **Song Library**.
 - **Page +**: Next **Song Library** page.
 - **Page -**: Previous **Song Library** page.
- **Song Jump**: Jump to a chosen **Song** (setlist or library context).
- **Tap Marker View** (tab) →
- **Tap Marker**: Start sending from this line for the matching tap type (see **Programming Multi-Tap**).
- **View Jump**: Switch to a chosen screen when the Preset runs.

Button Actions

Assigned in **Control** → controller → **Assign Action**. Tabs: **Main Actions** | **Song Actions** (page 1); **Focus** | **Navigate** | **Learn** (page 2). **Song Actions** is disabled in **Bank Mode**.

Main Actions

- **Preset** (category) →

- **Bank Preset** → **Select a Bank Preset (1st–10th, Default)**: slot in the current bank (button 1 → Preset 1 in bank 1, Preset 11 in bank 2, ...).
- **Global Preset**: one library Preset (1–120), same in every bank.
- **Preset** – / +: previous / next Preset in the bank.
- **Bank** (category) →
- **Bank** – / +: previous / next bank.
- **Bank Jump** → **Select a Bank** (1–12).
- **Bank Toggle** (1st Tap / Hold only) → **Select a Bank**: jump there; press again to return.
- **Bank A / B** (1st Tap / Hold only) → **Select Bank A and B**: flip between two banks.
- **Tap Tempo**: set **global BPM** from taps (or use a **Tap Tempo** Preset message).
- **Encoder Click** (1st Tap / Hold only): EC push-button action; use with **Encoder** message(s). Options include **Increase** / **Decrease** and **Active Preset** / **Global Preset** scope.
- **View** (category) →
- **View Jump** → e.g. **Preset Editor**, **Preset Library**, **Tracking**, **Live**, **Playing Song**, **Playing Setlist**, **Song Library**, **Setlist Library**, **XY Controller**, **EC Controller**, **Footswitch Pedal**, **Expression Pedal**, **MIDI In**.
- **View A/B** (1st Tap / Hold only) → **Select View A and B**.
- **View Toggle**: open a chosen screen or return to the previous one.

Song Actions

Song Mode only (**Assign Action** tab **Song Actions**).

- **Song Preset** (category) →
- ← → ↑ ↓: Move the highlight on **Playing Song** (scroll slots without a fixed **1st–10th** assignment).
- **Position: Position** picker — choose which Song Preset slot is active (same role as directional scroll).
- **1st–10th, Default (Select a Song Preset)**: Always fire that slot in the current **Song** (e.g. **3rd** = chorus on every **Song**).
- **Setlist Library** (category) →
- **Song(S)** – / **Song(S)** +: Previous / next **Song** in the **Playing Setlist**.
- **Setlist** – / **Setlist** +: Previous / next **Setlist** in **Setlist Library**.
- **Setlist Jump**: Jump to a chosen **Setlist** in **Setlist Library**.
- **Song Library** (category) →
- **Song(L)** – / **Song(L)** +: Previous / next **Song** in **Song Library** (all songs, not only the playing setlist).
- **Page** – / **Page** +: Previous / next **Song Library** page (not setlist pages).
- **Song Jump**: Jump to a chosen **Song** (setlist or library context when you assign it).

Navigate

(**Navigate** tab — focus on Song / Setlist performance screens.)

- **None**

- ↑ ↓ ← → — move the on-screen cursor.
- **Enter** — confirm selection.
- **Previous Screen** — back out of the current list (not the same as **View Toggle**).

Chapter E: Features & Specifications

Features

Graviton M1 / M2 is an all-in-one MIDI control system for professional pedalboards and recording studios.

Fast & Easy MIDI Programming

- Large **3.5-inch touchscreen LCD** for intuitive control
- **No PC editor required**
- Built-in SysEx library / editor on the touchscreen

Wireless Remote Control

- Controlled by **XY / EC / F10W / F6W Controllers** with fully programmable switches
- Supports MIDI messages, bank/song switching, and more
- **650 ft (200 m)** wireless range, **encrypted signal**
- Battery life: 6 months (FxW), 1 month (XY) or 7 days (EC)

Footswitch & Expression Input

- **M2 Only:** 6 × soft-touch built-in footswitches (silent and smooth)
- TRS Input Ports:
- **M1:** 2 × Footswitch/Expression
- **M2:** 4 × Footswitch/Expression
- Wireless:
- Up to **10 F10W/F6W footswitches**
- Up to **10 XY/EC Controllers**
- **All can work together simultaneously**

Specialized I/O (Built-in for M2 or RV Adapter for M1)

- **Relay Port:** Control up to 2 amp channel switches or pedal footswitch inputs
- **Control Voltage Port:** Control up to 2 analog CV/expression inputs

Modes & Memory

- **Bank Mode** and **Song Mode** (on-screen **SONG MODE** toggle)
- **120 Presets** across **12 Banks**
- **120 Songs** and **120 Setlists**
- **120 SysEx messages**
- **10 Expression Presets**
- **10 Backups** (each stores full user data: 120 Presets, Songs, Setlists, SysEx, 10 Expression Presets, Actions, Settings, etc.)

Message Types & Features

- 16 messages per Preset / Expression Preset
- **Unlimited Preset chaining**
- Supports:
 - Program Change, Control Change, Note On/Off
 - Counter PC/CC, SysEx
 - MIDI Clock (Start/Stop/Continue), Tap Tempo
 - Expression CC/PC, Clock BPM
 - Expression-triggered Presets (Toe/Heel Down)
 - Preset-triggered Expressions
 - And many more

MIDI Connectivity & Routing

- MIDI In / Out / Through (thru) / Clock routing
- Standard **5-pin MIDI In/Out**
- **1/4" TRS MIDI:**
 - M1: 2 ports
 - M2: 4 ports
 - Supports Type A, Type B, Tip Active, Ring Active
- **1/8" (3.5mm) TRS MIDI Out** — M1 **TRS 3** and M2 **TRS 5 (MIDI Out Only)** jacks; same four MIDI Out types)
- **1/8" (3.5mm) TRS MIDI In** — M2 only (Type A); use **either** this **or** 5-pin DIN MIDI In, not both
- **USB MIDI** (Bus-powered)
- **USB Host MIDI** (can power external devices)
- **Bluetooth LE MIDI**

Connectivity & Updates

- **WiFi-based user data Import/Export**
- **Over-the-Air software updates** via WiFi

Specifications

	Graviton M1	Graviton M2
LCD	3.5-inch IPS, wide viewing angle, sunlight readable	3.5-inch IPS, wide viewing angle, sunlight readable
Touch panel	Capacitive	Capacitive
Built-in footswitches	—	6 soft-touch
Power input	9VDC center negative / USB Type-C	9VDC center negative / USB Type-C
Power consumption	9V 250mA (9VDC) or 5V 500mA (USB bus powered)	9V 250mA (9VDC) or 5V 500mA (USB bus powered)
Powering via USB Host	9VDC powers both the M1 and the connected device	9VDC powers both the M2 and the connected device
Finish	Premium brushed stainless steel	Premium brushed stainless steel
W × D × H	3.74 × 3 × 1.38 in (95 × 76 × 35 mm)	5.9 × 4.52 × 2.4 in (150 × 115 × 63 mm)
Weight	9 oz (250 g)	22.5 oz (640 g)